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(54) **RELEASE OF GROWTH FACTORS AT WOUND HEALING STAGES**

(71) Applicants: **THE BOARD OF TRUSTEES OF THE UNIVERSITY OF ARKANSAS**, Little Rock, AR (US); **NATIONAL UNIVERSITY CORPORATION OKAYAMA UNIVERSITY**, Okayama (JP)

(72) Inventors: **Joshua Sakon**, Fayetteville, AR (US); **Robert Beitle**, Fayetteville, AR (US); **Hazim Aljewari**, Fayetteville, AR (US); **Osamu Matsushita**, Okayama (JP); **Stephanie Beitle**, Fayetteville, AR (US)

(73) Assignees: **THE BOARD OF TRUSTEES OF THE UNIVERSITY OF ARKANSAS**, Little Rock, AR (US); **NATIONAL UNIVERSITY CORPORATION OKAYAMA UNIVERSITY**, Okayama (JP)

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See application file for complete search history.

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Primary Examiner — Lianko G Garyu

Assistant Examiner — Mercy H Sabila

(74) *Attorney, Agent, or Firm* — Quarles & Brady, LLP

(57) **ABSTRACT**

The present invention provides collagen-binding agents that can be used to treat wounds, ischemic heart disease, and other conditions. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The present invention further provides pharmaceutical compositions and biomedical devices comprising the disclosed collagen-binding agents, as well as methods for treating a condition using the collagen-binding agents.

6 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

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Figure 1

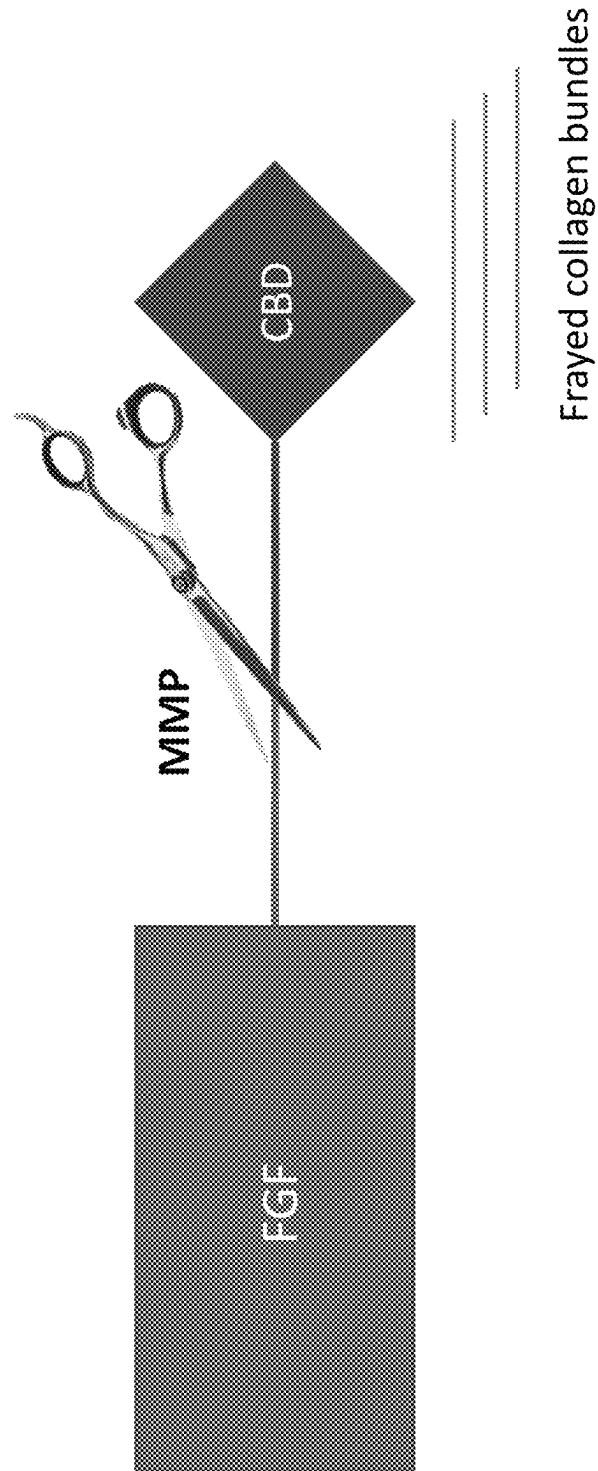


Figure 2

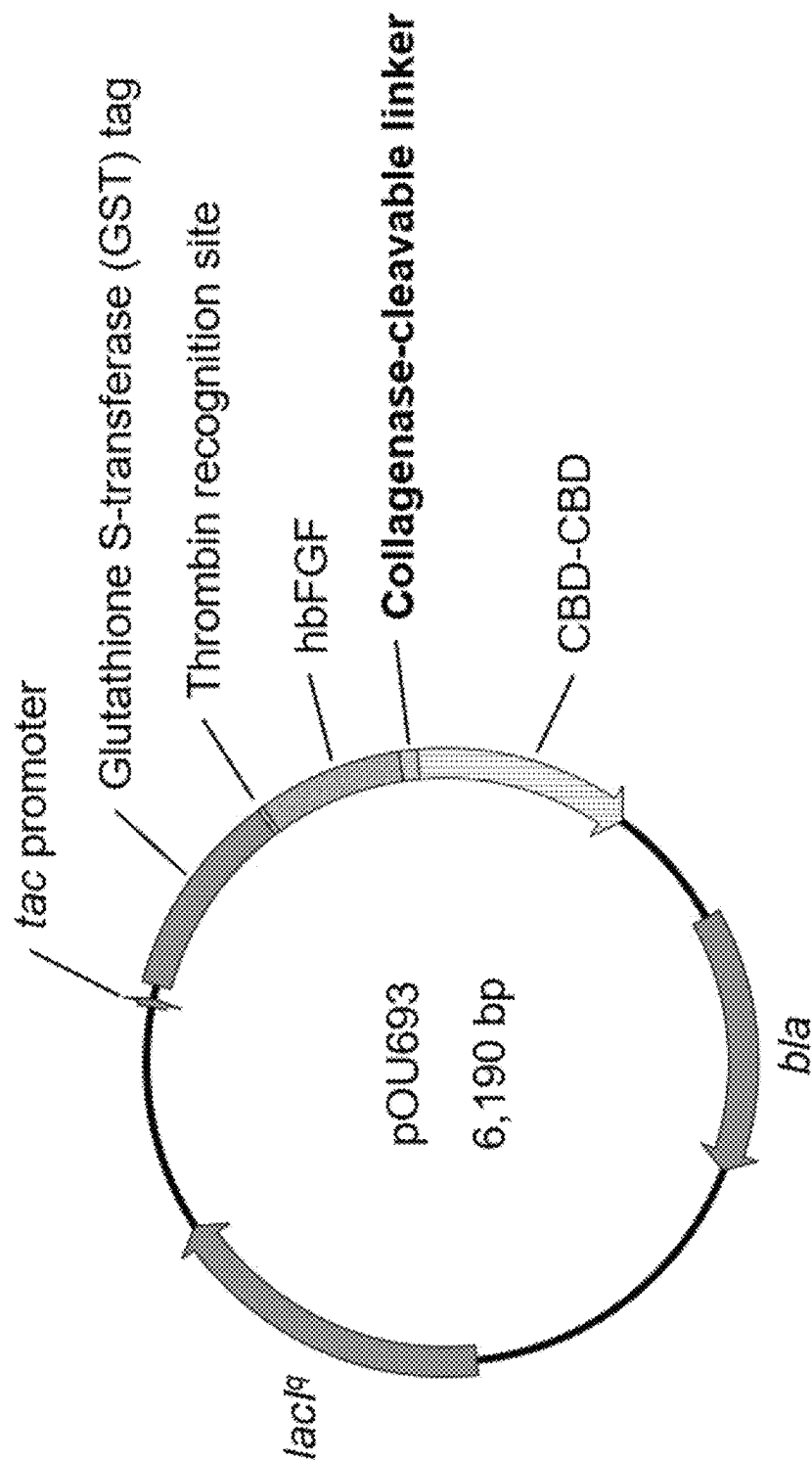


Figure 3

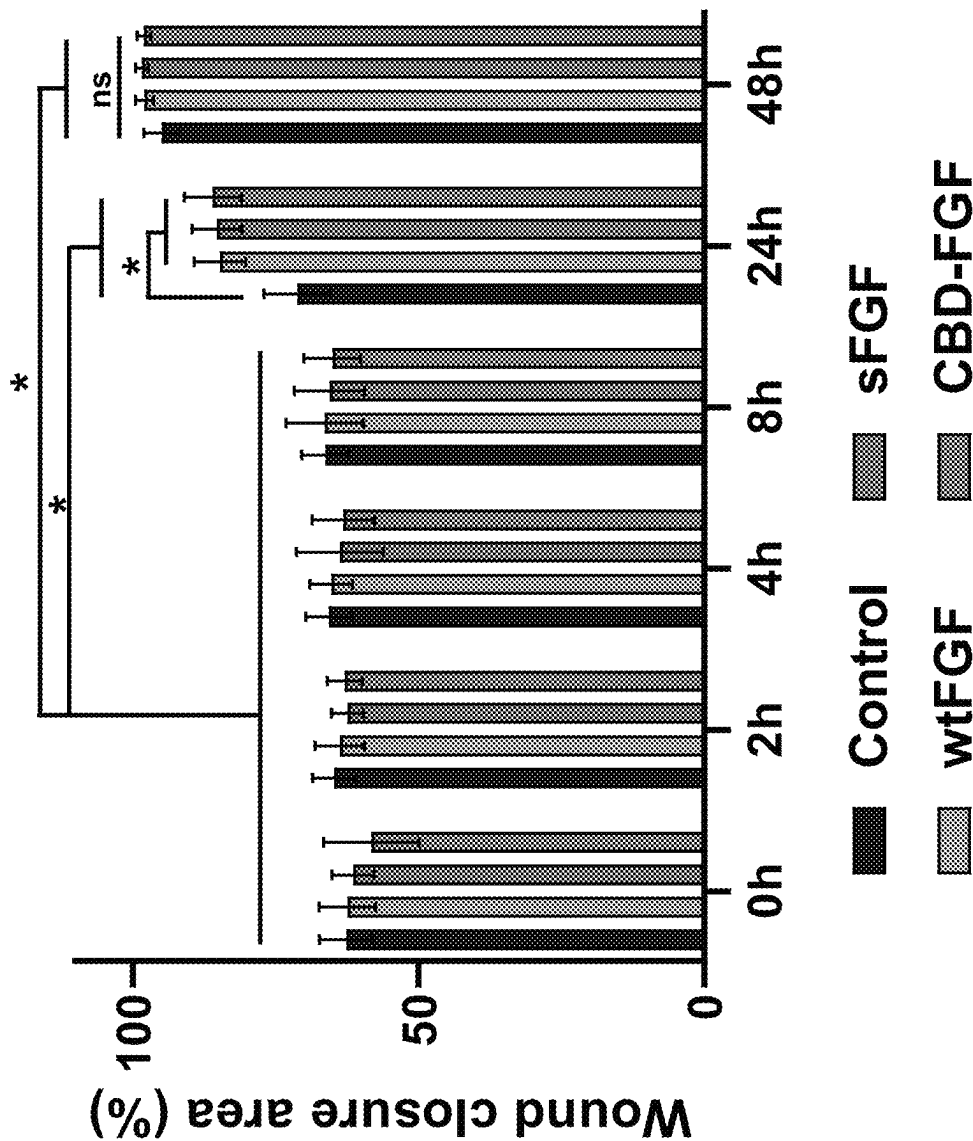


Figure 4

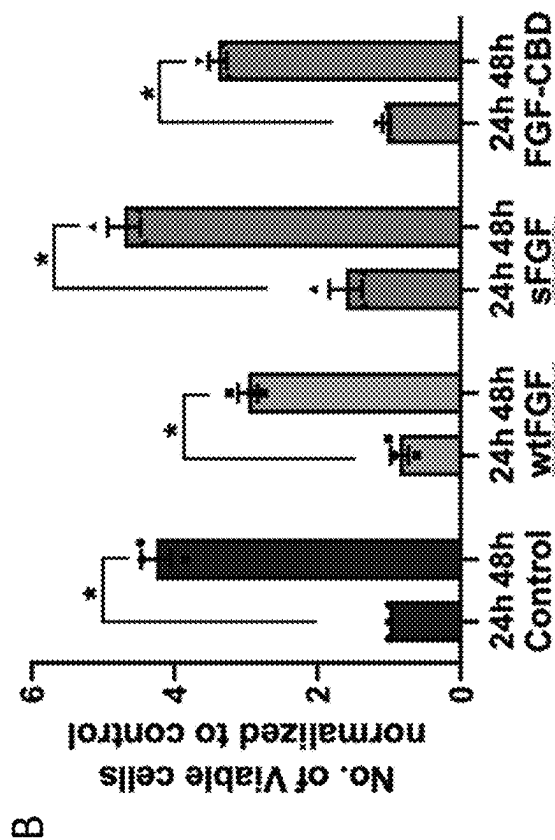
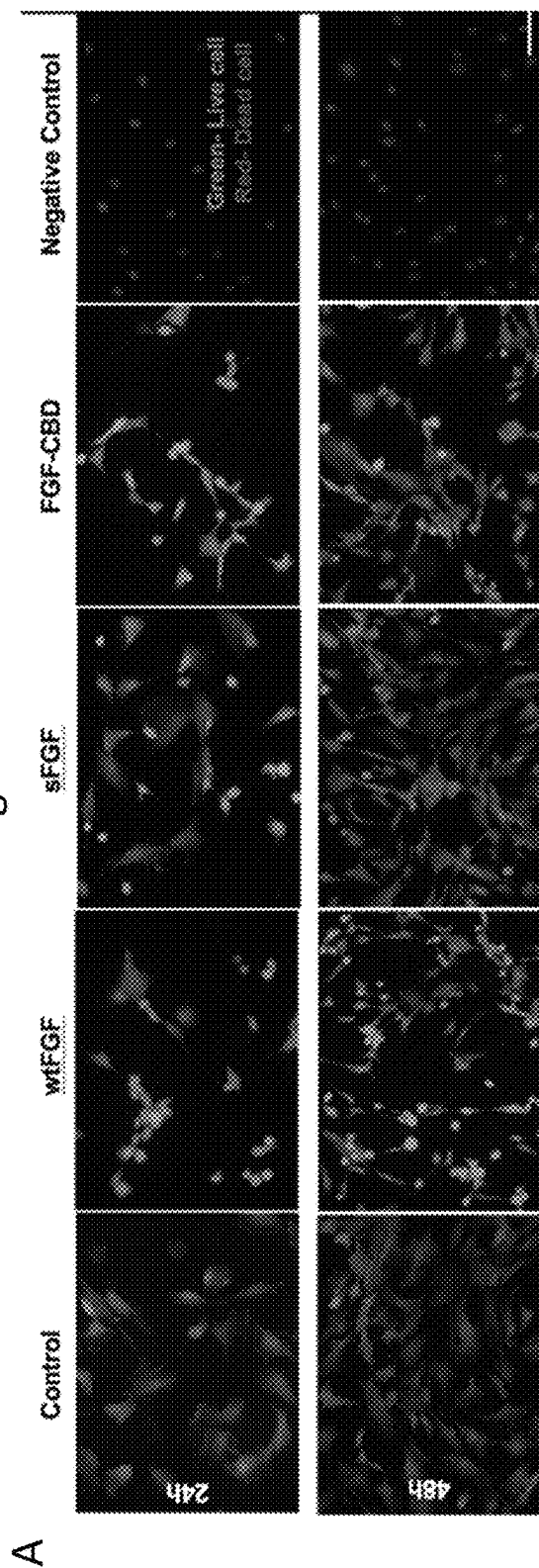


Figure 5

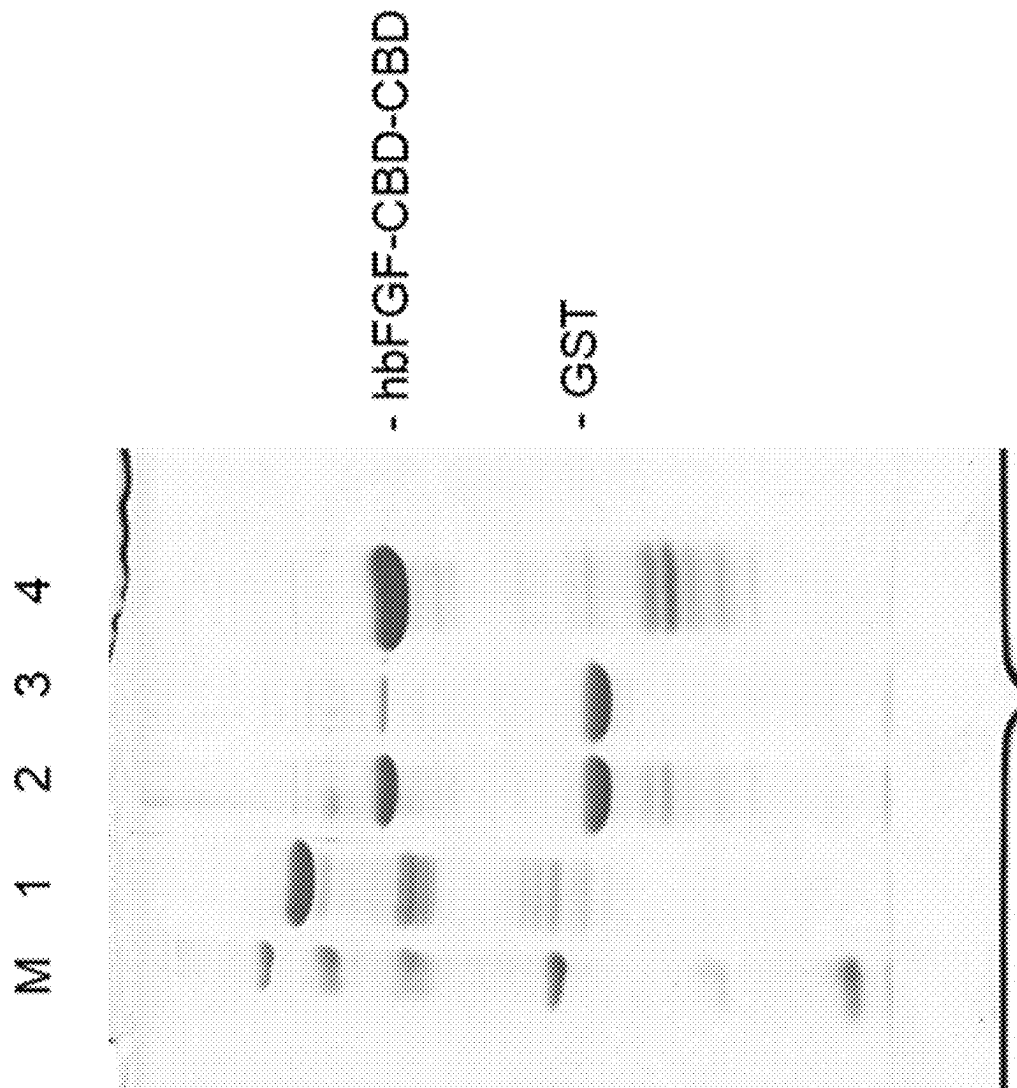


Figure 6

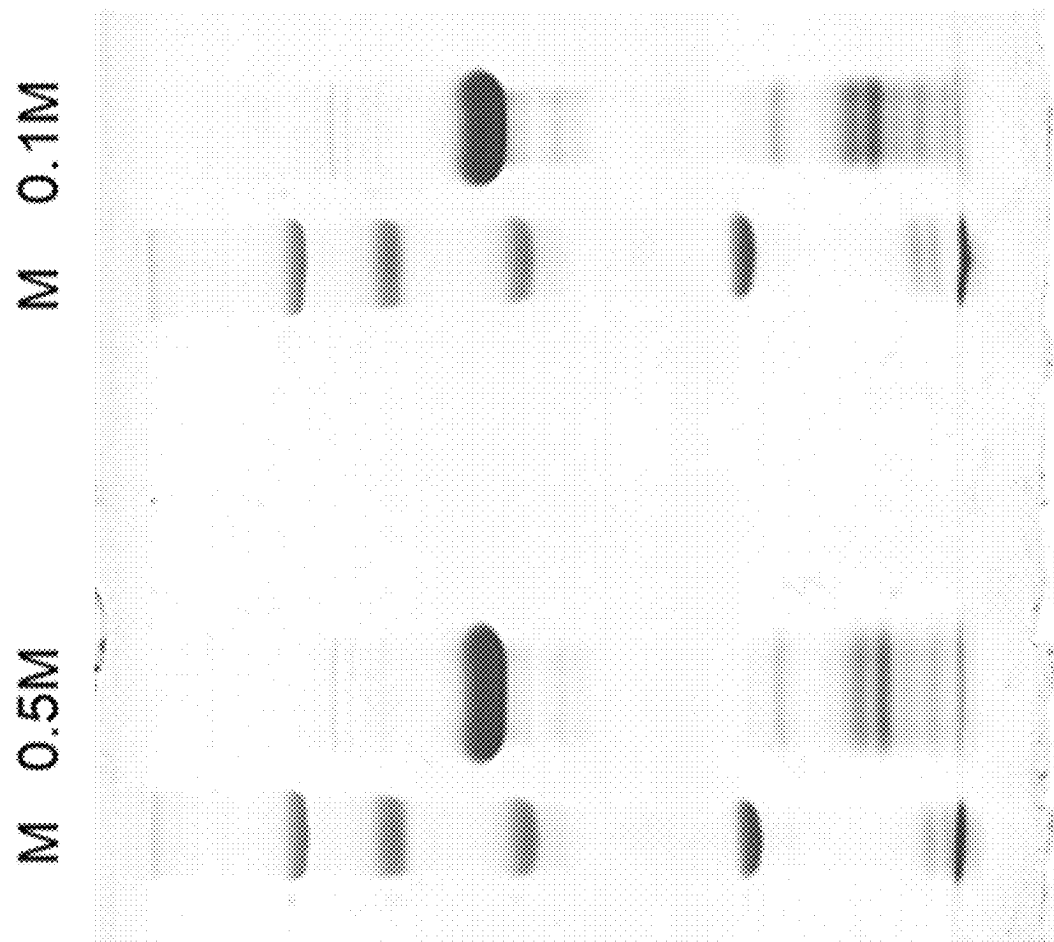
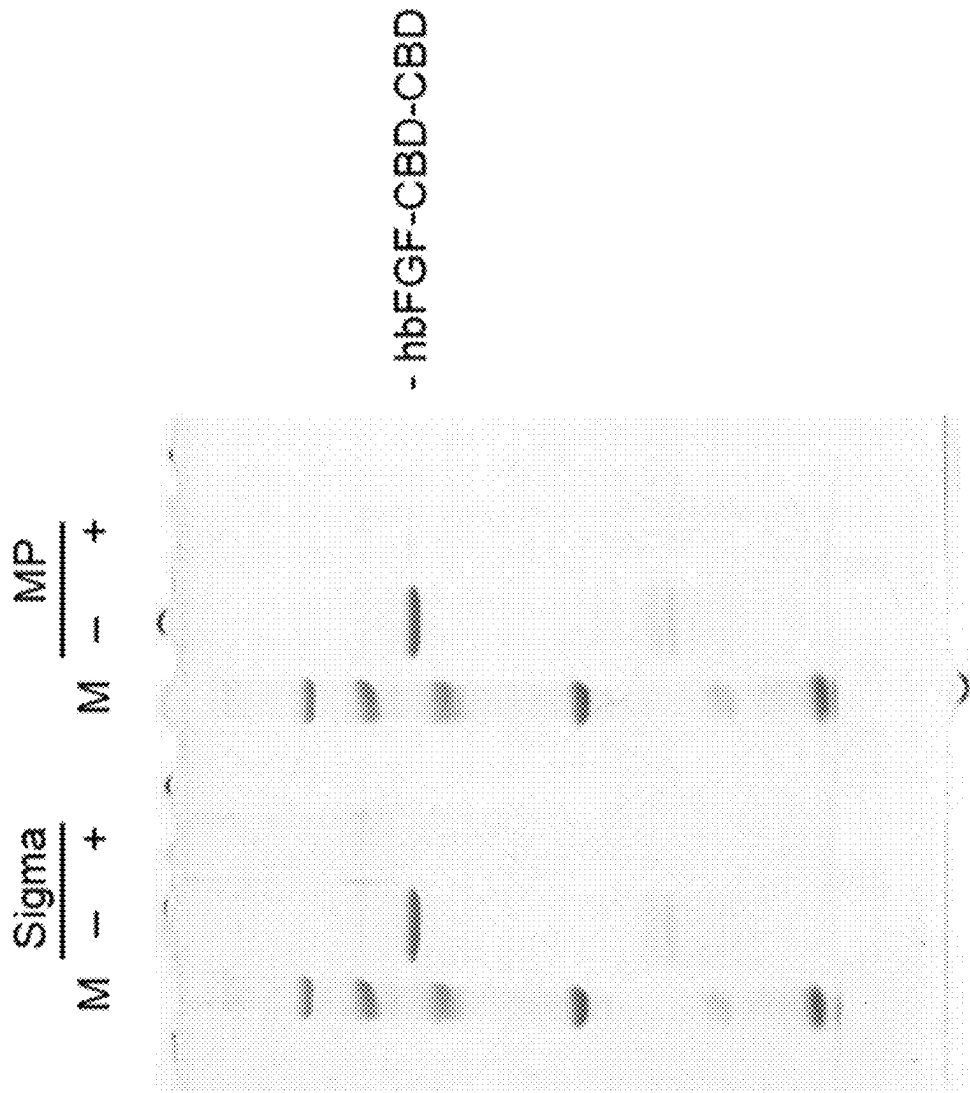


Figure 7



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RELEASE OF GROWTH FACTORS AT WOUND HEALING STAGES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/151,023 filed on Feb. 18, 2021, the contents of which are incorporated by reference in their entireties.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH

This invention was made with government support under grant numbers GM130174 and GM103429 awarded by the National Institutes of Health. The government has certain rights in this invention.

SEQUENCE LISTING

A Sequence Listing accompanies this application and is submitted as an ASCII text file of the sequence listing named "169879_00150_ST25.txt" which is 80,841 bytes in size and was created on Feb. 4, 2022. The sequence listing is electronically submitted via EFS-Web with the application and is incorporated herein by reference in its entirety.

BACKGROUND

Rapid diagnosis and percutaneous coronary intervention after a coronary blockage improve patient survival. Nonetheless, one in four patients develop heart failure within four years of a first heart attack. The American Heart Association reports that 635,000 Americans have a new myocardial infarction each year with 275,000 deaths attributable to heart failure (2009 data). Healthcare costs and productivity losses due to cardiovascular disease cost the U.S. \$273-\$320 billion annually according to the American Heart Association. Assurant Employee Benefits, an insurance provider, reported that the lifetime cost of treating less severe heart attacks is \$760,000 per person, which represents a substantial burden on heart patients and their families. For a severe heart attack, the cost can reach \$1 million.

One in four first-time heart attack victims will experience some measure of heart failure within four years of the initial episode. Currently, there are no therapeutics available that can promote regeneration of the heart. While available treatments reduce the symptoms of myocardial ischemia-induced injury and delay the onset of heart failure, they are unable to reverse cardiac damage. Accordingly, there remains a need in the art for improved therapeutics that promote regeneration of heart tissue.

SUMMARY

In a first aspect, the present invention provides collagen-binding agents comprising a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The collagen-binding domain is a polypeptide selected from the group consisting of SEQ ID NOs:6-53, a polypeptide having at least 90% sequence identity to SEQ ID NOs:6-53, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-53. In some embodiments, the therapeutic agent is a fibroblast growth factor (FGF) polypeptide. In some embodiments, the protease cleavage site is the cleavage site of a matrix metalloproteinase (MMP).

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In a second aspect, the present invention provides pharmaceutical compositions comprising the collagen-binding agents disclosed herein and a pharmaceutically acceptable carrier.

5 In a third aspect, the present invention provides biomedical devices comprising a coating that comprises a collagen-binding agent disclosed herein.

In a fourth aspect, the present invention provides methods of treating a condition. In a first embodiment, the methods comprise administering a collagen-binding agent or a pharmaceutical composition disclosed herein to a subject in an amount effective to treat the condition. In a second embodiment, the methods comprise administering a biomedical device disclosed herein to a subject in need of treatment with the therapeutic agent.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic of the collagen-binding fusion protein disclosed herein. The fusion protein comprises a fibroblast growth factor (FGF) protein fused to a collagen-binding domain (CBD) via a linker peptide that comprises a matrix metalloproteinases (MMP) cleavage site.

FIG. 2 is a vector map of the pCHG112-bFGF (pOU693) plasmid described in Example 3. This plasmid encodes a fusion protein in which human basic FGF (hbFGF) and a tandem CBD (CBD-CBD) are separated by a collagenase-cleavable linker such that these portions of the fusion protein can be cleaved apart by a collagenase. On the N-terminus, the fusion protein comprises (1) a glutathione S-transferase (GST) tag for affinity purification with glutathione-Sepharose beads, and (2) a thrombin recognition site that allows the GST tag to be removed via thrombin cleavage following affinity purification. Fusion protein expression is driven by the tac promoter. Inclusion of a lacI^q repressor makes this expression inducible via addition of isopropyl-β-D-thiogalactopyranoside (IPTG). Finally, the plasmid includes a bla gene, which encodes β-lactamase, making transformants resistant to ampicillin.

FIG. 3 is a bar graph showing the results of a cardiac fibroblast scratch assay in which the cardioprotective function of the bFGF-CBD-CBD fusion protein (CBD-FGF) was compared to that of wild-type FGF (wtFGF), super FGF (sFGF), and no FGF (control).

FIG. 4 demonstrates that the bFGF-CBD-CBD fusion protein does not negatively impact cell viability. A live/dead assay kit was used to quantify the percentage of live cells in cultures treated with wild-type FGF (wtFGF), super FGF (sFGF), the bFGF-CBD-CBD fusion protein (FGF-CBD), or no FGF (control) after 24 and 28 hours. Images of the cells were taken using a fluorescence microscope (A) and the number of live cells was quantified (B).

FIG. 5 is a Coomassie-stained SDS-PAGE gel demonstrating the purification of the hbFGF-CBC-CBD fusion protein. The fusion protein was produced as a GST-fusion protein (1). GST-tag was cleaved off using thrombin protease (4). The mixture was then further separated by Heparin-Sepharose column chromatography. Both the unbound (2) and bound (3) fractions were analyzed.

FIG. 6 is a Coomassie-stained 12.5% SDS-PAGE gel demonstrating the solubility of the purified hbFGF-CBC-CBD fusion protein. Following elution from the Heparin-Sepharose column, the fusion protein was first dialyzed against 50 mM TrisHCl (pH7.5), 0.5 M NaCl (7.4 μg, 0.5M), and was then dialyzed against 50 mM TrisHCl (pH7.5), 0.1 M NaCl (7.4 μg, 0.1M). M: 94, 67, 43, 30, 20.1, and 14.4 kDa markers. 0.5M: the fraction dialyzed against the buffer

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containing 0.5 M NaCl (7.4 μ g). 0.1M: the fraction dialyzed against the buffer containing 0.1 M NaCl (7.4 μ g).

FIG. 7 is a Coomassie-stained 12.5% SDS-PAGE gel demonstrating the ability of the hbFGF-CBC-CBD fusion protein to bind to collagen. 200 pmol of hbFGF-CBC-CBD was dissolved in 100 μ l of 50 mM TrisHCl (pH7.5), 0.1 M NaCl, 1 mM CaCl_2 , and mixed with 10 mg of collagen powder prewashed in the same buffer. After incubation for 30 minutes at 4° C., 15 μ l each of the unbound fractions were analyzed. M: 94, 67, 43, 30, 20.1, and 14.4 kDa markers. Sigma: Sigma collagen Type I (C-9879). MP: MP collagen (insoluble, Cat No 160083).

DETAILED DESCRIPTION

The present invention provides collagen-binding agents that can be used to treat wounds, ischemic heart disease, and other conditions. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The present invention further provides pharmaceutical compositions and biomedical devices comprising the disclosed collagen-binding agents, as well as methods for treating a condition using the collagen-binding agents.

Growth factors enhance regeneration. As a result, growth factor-based treatments that promote regeneration of heart tissue following a heart attack have long been desired. However, the challenges associated with targeted delivery of growth factors and off-target effects of the growth factors have severely limited their clinical use. To overcome these challenges, the collagen-binding agents of the present invention have been designed to be both spatially and temporally regulated. Inclusion of a collagen-binding domain causes the collagen-binding agents to be lesion-seeking, targeting them to exposed collagen fibrils. Further, the collagen-binding agents are designed such that upregulation of a protease that cleaves the protease cleavage site temporally controls the release of the therapeutic agent from the collagen-binding domain.

In some embodiments, the collagen-binding agents are fusion proteins. For instance, in Example 1, the inventors describe the generation of a fusion protein that comprises a non-mitogenic fibroblast growth factor (FGF) protein as a therapeutic agent. This protein was selected because it is known to accelerate fibroblast cell migration and is hypothesized to have cardioprotective effects and to promote cardiac regeneration. The inventors' fusion protein comprises an FGF protein linked to a collagen-binding domain via a linker peptide that comprises an MMP cleavage site. The MMP cleavage site allows FGF to be released from the fusion protein when the cognate MMP is upregulated during the inflammatory phase of the wound healing process. While FGF is known to be effective for the treatment of acute wounds, its clinical use has been severely limited by its ability to promote tumorigenesis. Thus, controlling the delivery of FGF, both spatially and temporally, will allow it to aid in a critical phase of wound healing, while reducing its off-target effects and improving its clinical safety.

The inventors envision that their collagen-binding fusion proteins could be used to promote heart regeneration following myocardial ischemia-induced injury. Specifically, they imagine that their fusion protein could be delivered within the proximity of the heart via inclusion in the coating of a stent. Because stent placement is part of the standard treatment regimen, administration of the fusion protein in this manner would not significantly change the typical clinical workflow. Initially, to avoid excess fibrin formation,

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the fusion protein is not released from the stent and remains inactive, reducing the risk for in-stent restenosis. Once the fusion protein is released from the stent, the collagen-binding domain will target it to sites of exposed collagen fibers that accumulate during post-myocardial infarction remodeling. MMP upregulation during adverse ventricular remodeling would then release FGF from the fusion protein, allowing it to aid in the remodeling of the fibrin clot into new extracellular matrix (ECM). Importantly, release of FGF would occur during the stage of wound healing at which FGF will be most beneficial to patients. Thus, this collagen-binding agent represents a promising means to promote heart tissue remodeling and prevent a heart failure.

Collagen-Binding Agents:

In a first aspect, the present invention provides therapeutic collagen-binding agents. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. Specifically, the collagen-binding domain is a polypeptide selected from the group consisting of SEQ ID NOs:6-53, a polypeptide having at least 90% sequence identity to SEQ ID NOs:6-53, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-53 capable of binding collagen.

Collagen-Binding Domain

The "collagen-binding domain (CBD)" is a polypeptide that binds to collagen. The ability of a polypeptide to bind to collagen can be assessed as described in U.S. Patent Publication No. 2010/0129341 or International Publication No. WO2008/124166, which are incorporated herein by reference in their entirety. Briefly, the polypeptide is incubated with collagen in a buffer, and the mixture is passed through a filter that allows for the passage of the polypeptide but blocks the passage of collagen, such that passage of the polypeptide is blocked by the filter only if it binds to collagen. The filtrate is then assayed for the presence of the polypeptide. Suitably, at least 80%, 85%, 90%, 95%, 98%, or 99% of the collagen-binding domain is retained by the filter in this assay as compared to when the filtration is performed in the absence of collagen.

The collagen-binding domains disclosed herein are segments of collagenase proteins found in bacteria. In some embodiments, the collagen-binding domain is derived from ColG, a class I collagenase from *Clostridium histolyticum* (J. Bacteriol. 181:923-933, 1999), or from ColH, a class II collagenase from *Clostridium histolyticum* (J. Bacteriol. 176: 6489-6496, 1994). Compositions comprising a collagen-binding domain from ColH are described in US Patent Publication No. 2010/0129341; International Publication No. WO2008/124166; International Publication No. WO2018/148573, each of which is hereby incorporated herein by reference in its entirety. In previous work, the inventors demonstrated that the C-terminal collagen-binding domain of the *Clostridium histolyticum* collagenases bind to partially untwisted or undertwisted regions of collagen. This work is described in U.S. Pat. No. 9,579,273, which is hereby incorporated herein by reference in its entirety. This affinity for unwound collagen allows these domains to be used to target the collagen-binding agents of the present invention to lesions, which are characterized by disrupted collagen fibers.

The collagen-binding domain may also be any one of the polypeptides provided as SEQ ID NOs:6-53, which include collagen-binding domains derived from collagenases from various bacterial species, i.e., *Clostridium* and *Bacillus* species. While the collagen-binding domains from different bacteria share a relatively small amount of sequence identity, they all bind to collagen in a similar fashion. Thus, any of

the collagen-binding domains disclosed herein, as well as variants and fragments thereof, may be used in the collagen-binding agents of the present invention.

The phrase “derived from” indicates that the collagen-binding domain is a fragment of the full-length collagenase protein, that it contains amino acid changes relative to the wild-type protein, or a combination thereof. A collagen-binding domain may be derived from any collagen-binding protein by selecting a portion of the protein that binds to collagen. It is only required that the collagen-binding domain retains the ability to bind collagen. Suitably, the collagen-binding domains used with the present invention lack collagenase activity. “Collagenase activity” refers to the ability of a polypeptide to degrade or breakdown collagen, which would be harmful to a subject. For example, extensive degradation of collagen in connective tissues results in gas gangrene. The collagenase-derived collagen-binding domains used with the present invention may lack collagenase activity either because they comprise only a portion of a full-length collagenase protein or because they contain a mutation that eliminates collagenase function.

In previous work, the inventors developed “tandem collagen-binding domain” proteins (abbreviated herein as CBD-CBD) that comprises two collagen-binding domains, allowing for tighter binding to collagen via bridging of two collagen fibrils. This work is described in US Patent Publication No. 2019/0376053, which is hereby incorporated herein by reference in its entirety. Because of their increased affinity for collagen, these tandem collagen-binding domain proteins may be more effective than their single-domain counterparts for certain applications, particularly for applications that require highly localized treatment. Thus, in some embodiments, the collagen-binding domain is a tandem collagen-binding domain. The tandem collagen-binding domain may include two collagen-binding domains of the same type or may include two collagen-binding domains of different types (i.e. two collagen-binding domains from two different origins). Furthermore, the two collagen-binding domains may be from the same bacterial species or from two different bacterial species. Each collagen-binding domain may be individually selected from the group consisting of SEQ ID NOs:6-45, a polypeptide having at least 90%, 92%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NOs:6-45, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-45 capable of binding to collagen. In some embodiments, the collagen-binding domain comprises a naturally occurring tandem collagen-binding domain. Suitable naturally occurring tandem collagen-binding domains include those disclosed as SEQ ID NO:46-53. For instance, the collagen-binding domain utilized in the fusion protein tested in the Examples is the tandem collagen-binding domain of SEQ ID NO:46, which comprises the s3a and s3b domains from the *C. histolyticum* collagenase ColG. Thus, in one embodiment, the collagen-binding domain comprises SEQ ID NO:46. The two collagen-binding domains within a tandem collagen-binding domain may be linked by a covalent bond and/or a linker or spacer moiety. In some embodiments, the two domains are linked via a native collagen-binding domain linker. Suitable native collagen-binding domain linkers include, but are not limited to, those disclosed as SEQ ID NO:54-61.

Therapeutic Agent

The collagen-binding agents of the present invention also comprise a therapeutic agent. The term “therapeutic agent” refers to a pharmaceutically active compound. Exemplary therapeutic agents include, without limitation, polypeptides, polynucleotides, small molecules, hormones, carbohydrates,

and lipids. In some embodiments, the therapeutic agent is selected from the group consisting of a fibroblast growth factor (FGF) polypeptide, parathyroid hormone (PTH), a PTH/parathyroid hormone-related peptide (PTHrP) receptor agonist, a PTH/PTHrP receptor antagonist, a bone morphogenic protein (BMP), granulocyte colony stimulating factor (G-CSF), an anti-sclerostin antibody, insulin-like growth factor 1 (IGF-1), vascular endothelial growth factor (VEGF), a transforming growth factor beta (TGF- β) protein, a TGF- β receptor, transforming growth factor alpha (TGF- α), a keratinocyte growth factor (KGF) protein, CT, a growth hormone (GH) protein, granulocyte-macrophage colony-stimulating factor (GM-CSF), epidermal growth factor (EGF), a platelet-derived growth factor (PDGF) protein, celiprolol, and connective tissue growth factor (CTGF). See U.S. Pat. Nos. 8,450,273; 9,579,273; and International Publication no. WO2018/148573, which are incorporated herein by reference.

In preferred embodiments, the therapeutic agent is a fibroblast growth factor (FGF) polypeptide. FGF polypeptides are a family of cell signaling proteins produced by macrophages that play an important role in various cellular processes like cell proliferation, migration, and differentiation. They are known to induce processes such as regeneration, morphogenesis, and angiogenesis. For example, FGF polypeptides are known to bind to heparin to increase the efficiency of mitogenic activity. In humans, 23 members of the FGF family have been identified. FGF1 is known to play a crucial role in wound healing and other significant clinical conditions. For example, FGF1 has been shown to promote nerve regeneration and angiogenic activity, which are critical for wound healing. FGF2, which is also referred to herein as basic fibroblast growth factor (bFGF), has been hypothesized to mediate the formation of new blood vessels (i.e., angiogenesis) during wound healing. Further, preliminary animal studies suggest that FGF2 protects the heart from injury associated with a heart attack, reducing tissue death and promoting improved function after reperfusion. Suitable FGF polypeptides for use with the present invention include those disclosed as SEQ ID NOs:2-5. In the Examples, the inventors utilized the wild-type FGF2 polypeptide of SEQ ID NO:4 as the therapeutic agent in their collagen-binding agent. Thus, in some embodiments, the FGF polypeptide comprises SEQ ID NO:4. The FGF1 and FGF2 polypeptides described herein may be full-length polypeptides (i.e., SEQ ID NOs:2-5) or may be functional fragments of a full-length FGF polypeptide.

In some embodiments, the FGF polypeptide is a wild-type FGF1 (SEQ ID NO:2) or FGF2 (SEQ ID NO:4) polypeptide. In other embodiments, the FGF polypeptide is hyper-stable variant FGF1 (SEQ ID NO:3) or FGF2 (SEQ ID NO:5) polypeptide. Although FGF1 and FGF2 polypeptides are promising therapeutics, they have low intrinsic stability and are highly susceptible to proteolytic degradation, especially by thrombin, which is usually present in abundance in the fibrin clots at the site of a wound. In previous work, the inventors developed hyper-stable variants of human FGF1 and FGF2 that are not only resistant to thrombin but also exhibit heparin-independent mitogenic/wound healing activity. This work is described in US Patent Publication No. 2019/0284252, which is hereby incorporated herein by reference in its entirety. Due to the extraordinary physical and bioactivity of these engineered variants, the present inventors have named the engineered human FGF1 variant (which comprises the mutations Q41P, S48L, H94S, K113N, and R123E) “super human acidic fibroblast growth factor 1 (shFGF1)” and have named the engineered human FGF2

variant (which comprises the mutations Q65L, N111S, K128N, and K138E) “super human acidic fibroblast growth factor 2 (shFGF2).” The inventors demonstrated that shFGF1 shows no signs of degradation even when stored at room temperature (25° C.) for over 3 months. Specifically, shFGF1 denatures only at temperatures higher than 80° C., and it exhibits a wider range of pH stability (4.0-10.0) than wild-type human FGF1. shFGF1 shows no binding affinity to heparin but its mitogenic activity is higher than that of wild-type FGF1. Importantly, the inventors have shown that shFGF1 activates pathways involving the anti-apoptotic protein kinases PI3K-Akt and MEK1/2-ERK1/2. Thus, treatment with shFGF1 is expected to attenuate lethal myocardial reperfusion injury and limit myocardial infarct size. Also encompassed are FGF proteins having at least 90%, 92%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the polypeptides of SEQ ID NO: 2-5.

Protease Cleavage Site

The collagen-binding agents of the present invention also comprise a protease cleavage site. In some embodiments, the protease cleavage site is the cleavage site of a matrix metalloproteinase (MMP). MMPs are a family of calcium-dependent, zinc-containing endopeptidases that degrade extracellular matrix proteins. Exemplary MMPs for use with the present invention include MMP-1, MMP-2, MMP-3, MMP-7, MMP-8, MMP-9, MMP-10, MMP-13, MMP-14, and MMP-18, which cleave various forms of collagen. Suitable MMP cleavage sites for inclusion in the collagen-binding agents include, without limitation, GPQGIA (SEQ ID NO:66), GPQGIL (SEQ ID NO:67), GPQGLA (SEQ ID NO:68), GPQGIL (SEQ ID NO:69), GPLGIA (SEQ ID NO:70), GPLGIL (SEQ ID NO:71), GPLGLA (SEQ ID NO:72), GPLGLL (SEQ ID NO:73), GPRGLQ (SEQ ID NO:74), and GPTGLA (SEQ ID NO:75). In the fusion protein tested in the Examples, the inventors utilized the cleavage sequence for MMP-1 (i.e., GPLGIAGP; SEQ ID NO:65). Thus, in some embodiments, the protease cleavage site comprises SEQ ID NO:65.

Wound healing occurs in four phases: hemostasis phase, inflammatory phase, proliferative phase, and maturation phase. At the end of the inflammatory phase, the expression of specific MMPs (i.e., MMP-2, MMP-8, MMP-9, and MMP-13 initially, then MMP-10 and MMP-14) is upregulated. Thus, including a cleavage site for one such MMP between the collagen-binding domain and the therapeutic agent in a collagen-binding agent allows the therapeutic agent to be released during this critical phase of wound healing. Because different MMPs are upregulated at slightly different points in the wound healing process, the MMP cleavage site included in the collagen-binding agent may be selected such that the therapeutic agent is released at the stage in which it will provide the most benefit. A composition comprising more than one fusion protein is also envisioned. The composition may comprise fusion proteins with various cleavage sites to allow for extended release of the therapeutic agent over the different healing phases based on the MMP upregulated at specific phases of healing.

Construction

The collagen-binding agents of the present invention may be constructed in several ways. In some embodiments, the collagen-binding agent is a fusion protein, and the therapeutic agent is a polypeptide that forms one functional segment of the fusion protein. The term “fusion protein” refers to a single polypeptide comprising at least two functional segments. For example, the fusion proteins of the present invention comprise a collagen-binding segment comprising the collagen-binding domain, a protease cleav-

age site, and a therapeutic segment comprising the therapeutic agent. Each polypeptide segment may comprise a synthetic polypeptide, a naturally occurring polypeptide, a fragment of a naturally occurring polypeptide, or a variant polypeptide comprising one or more mutations. The polypeptide segments of the fusion protein can be linked together directly (e.g., via a peptide bond or chemical cross-linking) or indirectly (e.g., via a polypeptide linker).

In some embodiments, the collagen-binding domain, protease cleavage site, and therapeutic agent are conjugated via polypeptide linkers (i.e., polypeptides that bridge two protein segments). The polypeptide linkers may be any length and may include traditional or non-traditional amino acids. For example, the peptide linker may be 1-100 amino acids long, and is suitably 5, 10, 15, 20, 25 or more amino acids long. The linker may “flexible” such that it has no required fixed structure in solution and the adjacent protein segments are free to move relative to one another, e.g., allowing the collagen-binding segment to bind to collagen without steric hindrance from the therapeutic segment. Preferred amino acid residues for flexible linker sequences include glycine, alanine, serine, threonine, lysine, arginine, glutamine, and glutamic acid, but are not limited thereto. In some embodiments, the linker is a linker that is natively found in a collagen-binding domain, e.g., the linkers of SEQ ID NOs: 54-61. In some embodiments, the polypeptide linker is a polycystic kidney disease domain derived from a bacterial collagenase. “Polycystic kidney disease (PKD)” domains comprise an Ig-like fold consisting of a beta-sandwich of seven strands in two sheets with a Greek key topology. In some embodiments, the PKD domain comprises SEQ ID NO:62 or SEQ ID NO:63.

In other embodiments, the therapeutic agent is linked to a polypeptide comprising the protease cleavage site and the collagen-binding domain via chemical cross-linking. In embodiments in which the therapeutic agent is a polypeptide, it may be cross-linked through an amino group by a reagent such as disuccinimidyl glutarate or glutaraldehyde. It may also be cross-linked through an amino group by derivatizing one polypeptide with SANH (succinimidyl-4-hydrazinonicotinate acetone hydrazone) and the other with SFB (succinimidyl-4-formyl benzoate), and then mixing the two derivatized polypeptides. Two polypeptides can be cross-linked between an amino group of one polypeptide and a carboxyl group of the other polypeptide via reaction with EDC (1-ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride). Other suitable cross-linking methods are described in U.S. Patent Application Publication Nos. 2006/0258569 and 2007/0224119.

In other embodiments, the therapeutic agent is linked to the protease cleavage site and collagen-binding domain using a tag system, i.e., a pair of agents that bind to each other with high affinity. Suitable tag systems include, without limitation, biotin/avidin, biotin/streptavidin, and digoxigenin (DIG) systems.

In some embodiments, the collagen-binding agent is produced as a fusion protein that further comprises a purification tag, i.e., a moiety that facilitates isolation of the fusion protein. Suitable purification tags include, but are not limited to, histidine (His), hemagglutinin (HA), cMyc, glutathione S-transferase (GST), Flag, V5, and NE tags. In preferred embodiments, the purification tag is a His tag or a GST tag. In some embodiments, the purification tag is removed from the collagen-binding agent after it has been purified. This can be accomplished by including a protease cleavage site between the purification tag and the collagen-binding agent within the fusion protein and treating the

fusion protein with the cognate protease following purification. For instance, in Example 3, the inventors included a thrombin cleavage site in their fusion protein to allow a GST tag to be removed via treatment with thrombin following affinity purification with glutathione-Sepharose beads. However, any protease cleavage site may be used for this purpose.

In the Examples, the inventors generated a fusion polypeptide comprising, from N-terminus to C-terminus: an FGF polypeptide, an MMP cleavage site, and a collagen-binding domain. Thus, in some embodiments, the C-terminus of the therapeutic agent is linked to the N-terminus of the protease cleavage site, and the C-terminus of the protease cleavage site is linked to the N-terminus of the collagen-binding domain, as depicted in FIG. 1. In certain embodiments in which the collagen-binding agent further comprises a PKD domain, the N-terminus of the PKD is linked to the C-terminus of the protease cleavage site and the C-terminus of the PKD is linked to the N-terminus of the collagen-binding domain.

In some embodiments, the fusion polypeptide is that of SEQ ID NO:1 (i.e., the fusion protein tested in the Examples), which comprises from N-terminus to C-terminus: wild-type FGF2 (SEQ ID NO:4), the cleavage sequence for MMP-1 (SEQ ID NO:65), and a natural tandem collagen-binding domain comprising the s3a and s3b domains of the *C. histolyticum* collagenase ColG (SEQ ID NO:46).

Definitions

The terms “polypeptide,” “protein,” and “peptide” are used interchangeably herein to refer to a series of amino acid residues connected by peptide bonds between the alpha-amino and carboxy groups of adjacent residues, forming a polymer of amino acids. Polypeptides may include modified amino acids. Suitable polypeptide modifications include, but are not limited to, acylation, acetylation, formylation, lipoylation, myristoylation, palmitoylation, alkylation, isoprenylation, prenylation, amidation at C-terminus, glycosylation, glycation, polysialylation, glypiation, and phosphorylation. Polypeptides may also include amino acid analogs. The terms “protein” and “polypeptide” are often used in reference to relatively large polypeptides, whereas the term “peptide” is often used in reference to small polypeptides, but usage of these terms in the art overlaps.

The polypeptides provided herein include fragments of full-length polypeptides. For example, the collagen-binding domains used with the present invention may comprise fragments of SEQ ID NOs:6-53. As used herein, a “fragment” is a portion of an amino acid sequence which is identical in sequence to, but shorter in length than, a reference polypeptide. A fragment may comprise at least 8, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, or more contiguous amino acid residues of a reference polypeptide. Thus, the collagen-binding domain fragments of the present invention may comprise or consist essentially of a contiguous portion of a full-length collagen-binding domain (i.e., SEQ ID NOs:6-53). Fragments may be preferentially selected from certain regions of a molecule. A fragment may include an N-terminal truncation, a C-terminal truncation, or both truncations relative to the full-length collagen-binding domain or agent. The N-terminal and/or C-terminal truncations may include removal of 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid residues from a reference polypeptide. Suitably, the polypeptide fragments retain at least 20%, 40%, 60%, 80%, or 100% of the biological activity of the reference polypeptide (e.g., collagen-binding, wound healing activity).

The polypeptides provided herein also include variant polypeptides, i.e., engineered polypeptides that comprise substitution mutations relative to a wild-type reference polypeptide. For example, the FGF1 variant shFGF1 comprises the mutations Q41P, S48L, H94S, K113N, and R123E relative to the wild-type FGF1 protein, and the FGF2 variant shFGF2 comprises the mutations Q65L, N111S, K128N, and K138E relative to the wild-type FGF2 protein. A variant may comprise one or more insertions, deletions, or substitutions of an amino acid residue relative to a wild-type reference molecule. An “insertion” refers to a change in an amino acid sequence resulting in the addition of one or more amino acid residues. An insertion may add 1, 2, 3, 4, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, or more amino acid residues. A “deletion” refers to a change in the amino acid sequence that results in the absence of one or more amino acid residues. A deletion may remove 1, 2, 3, 4, 5, 10, 20, 50, 100, 200, or more amino acids residues. A “substitution” refers to a change in an amino acid sequence in which one amino acid is replaced with a different amino acid due to a point mutation. The amino acid substitution may be a conservative replacement (i.e., a replacement with an amino acid that has similar properties) or a radical replacement (i.e., a replacement with an amino acid that has different properties). Suitably, the variant polypeptides retain at least 20%, 40%, 60%, 80%, or 100% of the biological activity of the reference polypeptide (e.g., collagen-binding, wound healing activity).

“Percentage of sequence identity” or “percentage of sequence similarity” is determined by comparing two optimally aligned sequences over a comparison window. The aligned sequences may comprise additions or deletions (i.e., gaps) relative to each other for optimal alignment. The percentage is calculated by determining the number of matched positions at which an identical nucleic acid base or amino acid residue occurs in both sequences, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Protein and nucleic acid sequence identities are evaluated using the Basic Local Alignment Search Tool (“BLAST”), which is well known in the art (*Proc. Natl. Acad. Sci. USA* (1990) 87: 2267-2268; *Nucl. Acids Res.* (1997) 25: 3389-3402). The BLAST programs identify homologous sequences by identifying similar segments, which are referred to herein as “high-scoring segment pairs”, between a query amino acid or nucleic acid sequence and a test sequence which is preferably obtained from a protein or nucleic acid sequence database. Preferably, the statistical significance of a high-scoring segment pair is evaluated using the statistical significance formula (*Proc. Natl. Acad. Sci. USA* (1990) 87: 2267-2268), the disclosure of which is incorporated by reference in its entirety. The BLAST programs can be used with the default parameters or with modified parameters provided by the user. For example, the variant collagen-binding domains used with the present invention may have at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% 98%, or 99% sequence identity to any one of SEQ ID NOs:6-53. Similarly, the therapeutic agents used herein may be variants of the wild-type sequences and have at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% 98%, or 99% sequence identity to any one of SEQ ID NO: 2-5 or any of the sequences encoding the therapeutic agents described herein.

Pharmaceutical Compositions:

In a second aspect, the present invention provides pharmaceutical compositions comprising a collagen-binding

agent disclosed herein. The pharmaceutical compositions may include a pharmaceutically acceptable carrier, excipient, or diluent that is nontoxic to the cell or animal being exposed thereto at the dosages and concentrations employed. Often a pharmaceutical agent is in an aqueous pH buffered solution. Pharmaceutically acceptable carriers are known in the art and include, but are not limited to, diluents (e.g., Tris-HCl, acetate, phosphate), preservatives (e.g., Thimerosal, benzyl alcohol, parabens), solubilizing agents (e.g., glycerol, polyethylene glycerol), emulsifiers, liposomes, nanoparticles, and adjuvants. Pharmaceutically acceptable carriers may be aqueous or non-aqueous solutions, suspensions, or emulsions. Examples of nonaqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Aqueous carriers include isotonic solutions, alcoholic/aqueous solutions, emulsions, or suspensions, including saline and buffered media.

In some embodiments, the pharmaceutical compositions are formulated for topical administration. For example, the pharmaceutical compositions may be formulated as gels, creams, or liposome preparations suitable for topical delivery. The pharmaceutical compositions may be formulated for delivery to the lower layers of the skin or facilitate extended release of the pharmaceutical at the site of application.

Biomedical Devices:

In a third aspect, the present invention provides biomedical devices comprising a coating that comprises a collagen-binding agent disclosed herein. The term "biomedical device" refers to a device that is administered to an animal for a medical purpose, e.g., for the diagnosis, treatment, or prevention of a disease or condition. Exemplary biomedical devices include, but are not limited to, stents, artificial ureters, diaphragms, intrauterine devices, heart valves, catheters, denture liners, prosthetic devices, ophthalmic lens applications, and artificial skin. In certain preferred embodiments, the biomedical device is a stent.

The collagen-binding agent may be coated onto the biomedical device using any suitable means known in the art. For example, the biomedical device may be coated with a material to which the collagen-binding agent has an affinity (e.g., collagen) or may be coated with a material to which the collagen-binding agent may be chemically linked (e.g., a biodegradable polymer). The collagen used to coat the biomedical device may be a synthetic collagen or a natural collagen of any type (i.e., type I, type II, type III, or type IV). Suitably, the collagen is from the same species as the subject to which the biomedical device is to be administered. Suitable biodegradable polymers that can be used to coat the biomedical device include, without limitation, polyphosphazenes, polyanhydrides, polyacetals, poly(ortho esters), polyphosphoesters, polycaprolactone, polyurethanes, polylactide, polycarbonates, and polyamides.

Methods:

In a fourth aspect, the present invention provides methods of treating a condition. In a first embodiment, the methods comprise administering a collagen-binding agent or a pharmaceutical composition disclosed herein to a subject in an amount effective to treat the condition. In a second embodiment, the methods comprise administering a biomedical device disclosed herein to subject in need of treatment with the therapeutic agent.

The methods of the present invention can be used to target any therapeutic agent to sites of exposed collagen. Thus, the methods may be useful for the treatment of a variety of conditions including, without limitation, a wound, a bone

condition, a spinal fusion, an ischemic heart disease, a peripheral nerve disorder, a spinal cord injury, and a kidney disease.

As used herein, "treating" or "treatment" describes the management and care of a subject for the purpose of combating a disease, condition, or disorder. Treating includes the administration of a collagen-binding agent, pharmaceutical composition, or biomedical device disclosed herein to prevent the onset of the symptoms or complications, to alleviate the symptoms or complications, to decrease recovery time, or to eliminate the disease, condition, or disorder.

The collagen-binding agents, pharmaceutical compositions, and biomedical devices described herein may be administered by any means known to those skilled in the art. As used herein, the terms "administering" and "administration" refer to the introduction of a substance into a subject's body. Such methods include, but are not limited to, oral administration, transdermal administration, administration by inhalation, nasal administration, topical administration, intravaginal administration, ophthalmic administration, intraaural administration, intracerebral administration, rectal administration, sublingual administration, buccal administration, and parenteral administration, including injectable such as intravenous administration, intra-arterial administration, intramuscular administration, intradermal administration, intrathecal administration, and subcutaneous administration. Administration can be continuous or intermittent.

The biomedical devices should be administered by a method that is appropriate in view of the particular device and condition to be treated. In some embodiments, the biomedical device is implanted into the subject. In particular embodiments, the biomedical device is implanted into an artery of the subject. In other embodiments, the biomedical device is applied to the skin or a tissue of the subject (e.g., to skin or tissue comprising a wound).

The collagen-binding agents and pharmaceutical compositions can be administered as a single dose or as divided doses. For example, the composition may be administered two or more times separated by 4 hours, 6 hours, 8 hours, 12 hours, a day, two days, three days, four days, one week, two weeks, or by three or more weeks. Optionally, such treatment may be repeated, for example, every 1, 2, 3, 4, 5, 6, or 7 days, or every 1, 2, 3, 4, and 5 weeks or every 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months.

The term "amount effective to treat the condition" refers to an amount that produces desirable biological or clinical results, e.g., reducing, alleviating, inhibiting, or preventing one or more symptoms of the condition. In some embodiments, the effective amount is an amount suitable to promote wound healing or provide a cardioprotective effect.

It will be appreciated that the specific dosage of the collagen-binding agent that is administered in any given case will be adjusted in accordance with the composition(s) being administered, the condition to be treated, the condition of the subject, and other relevant medical factors that may modify the activity of the compositions or the response of the subject. For example, the specific dose for a particular subject depends on age, body weight, general state of health, diet, the timing and mode of administration, the rate of excretion, medicaments used in combination and the severity of the particular condition to which the therapy is applied. Dosages for a given patient can be determined using a conventional pharmacological protocol. For example, in embodiments that utilize FGF1 or FGF2 polypeptides as a therapeutic agent, suitable dosage ranges may be on the order of several hundred micrograms of effective ingredient

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with a range from about 0.01 to 10 mg/kg/day, preferably in the range from about 0.1 to 1 mg/kg/day.

In some embodiments, the collagen-binding agent is co-administered with another therapeutic. Suitable therapeutics for co-administration with the collagen-binding agent include, without limitation, anti-platelet therapeutics (e.g., targeting P2Y12, GPIIb/IIIa or COX1), anti-coagulant therapeutics (e.g., enzyme replacement therapy, heparin/heparin mimic), cytotoxic therapeutics to control restenosis, hypertension therapeutics targeting RAS, or stem cell therapeutics.

The “subject” to which the methods are applied may be a mammal or a non-mammalian animal, such as a bird. Suitable mammals include, but are not limited to, humans, cows, horses, sheep, pigs, goats, rabbits, dogs, cats, bats, mice, and rats. In certain embodiments, the methods may be performed on lab animals (e.g., mice and rats) for research purposes. In other embodiments, the methods are used to treat commercially important farm animals (e.g., cows, horses, pigs, rabbits, goats, sheep, and chickens) or companion animals (e.g., cats and dogs). In a preferred embodiment, the subject is a human.

The present disclosure is not limited to the specific details of construction, arrangement of components, or method steps set forth herein. The compositions and methods disclosed herein are capable of being made, practiced, used, carried out and/or formed in various ways that will be apparent to one of skill in the art in light of the disclosure that follows. The phraseology and terminology used herein is for the purpose of description only and should not be regarded as limiting to the scope of the claims. Ordinal indicators, such as first, second, and third, as used in the description and the claims to refer to various structures or method steps, are not meant to be construed to indicate any specific structures or steps, or any particular order or configuration to such structures or steps. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to facilitate the disclosure and does not imply any limitation on the scope of the disclosure unless otherwise claimed. No language in the specification, and no structures shown in the drawings, should be construed as indicating that any non-claimed element is essential to the practice of the disclosed subject matter. The use herein of the terms “including,” “comprising,” or “having,” and variations thereof, is meant to encompass the elements listed thereafter and equivalents thereof, as well as additional elements. Embodiments recited as “including,” “comprising,” or “having” certain elements are also contemplated as “consisting essentially of” and “consisting of” those certain elements.

Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if a concentration range is stated as 1% to 50%, it is intended that values such as 2% to 40%, 10% to 30%, or 1% to 3%, etc., are expressly enumerated in this specification. These are only examples of what is specifically intended, and all possible combinations of numerical values between and including the lowest value and the highest value enumerated are to be considered to be expressly stated in this disclosure. Use of the word “about” to describe a particular recited amount or range of amounts is meant to indicate that values very near to the recited

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amount are included in that amount, such as values that could or naturally would be accounted for due to manufacturing tolerances, instrument and human error in forming measurements, and the like. All percentages referring to amounts are by weight unless indicated otherwise.

No admission is made that any reference, including any non-patent or patent document cited in this specification, constitutes prior art. In particular, it will be understood that, unless otherwise stated, reference to any document herein does not constitute an admission that any of these documents forms part of the common general knowledge in the art in the United States or in any other country. Any discussion of the references states what their authors assert, and the applicant reserves the right to challenge the accuracy and pertinence of any of the documents cited herein. All references cited herein are fully incorporated by reference, unless explicitly indicated otherwise. The present disclosure shall control in the event there are any disparities between any definitions and/or description found in the cited references.

The following examples are meant only to be illustrative and are not meant as limitations on the scope of the invention or of the appended claims.

EXAMPLES

Example 1: Generation of the bFGF-CBD-CBD Fusion Protein

In the following Example, the inventors describe the expression and purification of a collagen-binding fusion protein that comprises a human basic fibroblast growth factor polypeptide (bFGF; also known as FGF2) fused to a tandem collagen-binding domain polypeptide (CBD-CBD) via a linker peptide that comprises the cleavage site of matrix metalloproteinase-1 (MMP-1), such that these the bFGF and CBD-CBD portions of the fusion protein can be cleaved apart by MMP-1. See FIG. 1 and SEQ ID NO: 1.

Strain and Plasmid Construction
BL21(DE3) competent *E. coli* were purchased from New England Biolabs Inc. and used as a host to produce bFGF-CBD-CBD. The pBAD/Myc-His plasmid was purchased from Invitrogen Life Technologies. This plasmid contains the araBAD promoter and the araC gene. The araC gene encodes a protein that regulates the araBAD promoter, resulting in tight, dose-dependent, L-arabinose-inducible regulation of heterologous gene expression. The plasmid also contains a gene encoding lactamase, which provides ampicillin resistance and allows for selection of transformants that harbor the plasmid using this antibiotic. The DNA sequence encoding the bFGF-CBD-CBD protein (which is 424 amino acids in length) was purchased from IDT and cloned into the pBAD/Myc-His plasmid downstream of a sequence encoding a polyhistidine tag (His-tag).

Protein Expression
A single colony of BL21 *E. coli* comprising the pBAD bFGF-CBD-CBD plasmid was selected from a plate and inoculated into a 50 ml sterile tube containing 10 ml of LB media supplemented with 75 µg/ml ampicillin. The culture was incubated at 37° C. with shaking at 250 rpm overnight. L-arabinose was used to induce expression of His-tagged bFGF-CBD-CBD from the plasmid at the mid-exponential phase of growth.

Cell Lysate Preparation

Cells were collected by centrifugation at 4° C., 4,500 rpm for 45 minutes. The bacterial pellet was suspended by

adding 5 ml of 10 mM phosphate buffer supplemented with 0.50 of a protease inhibitor cocktail. Sonication was used to lyse cells.

Protein Purification

Cell lysates were filtered through a 0.20 μ m filter. A HiTrap IMAC FF column from GE Healthcare-Bio-Sciences (Uppsala, Sweden) was equilibrated with 0.2 M nickel and 10 mM sodium phosphate buffer (pH 7.3). His-tagged bFGF-CBD-CBD was eluted from the column using 250 mM imidazole in 10 mM sodium phosphate buffer (pH=7.3).

Expression Analysis

Expression of bFGF-CBD-CBD was confirmed by running the purified sample in a polyacrylamide (SDS-PAGE) gel and staining for His-tagged protein using a Pierce™ 6xHis Protein Tag Stain Reagent Set purchased from Thermo Fisher Scientific. Expression of bFGF-CBD-CBD was also confirmed by western blot using anti-His-tag antibodies.

Example 2: Assessment of the bFGF-CBD-CBD Fusion Protein

The bFGF-CBD-CBD fusion protein was first subjected to efficacy tests in vitro. First, cardioprotective function will be assessed using a cardiac fibroblast scratch assay [1]. Briefly, rat ventricular fibroblasts were isolated from 2-day old neonatal rat pups, seeded in 12-well plates, and grown in 2% serum medium. Two days after seeding, the cell monolayer was scraped in a straight line with a p200 pipet tip. Cell debris was removed by washing the cells once with 1 mL of medium supplemented with bFGF-CBD-CBD, super FGF, or no FGF. Samples were imaged every 6 hours for the next 24 hours and the degree of wound closure stimulated by the various treatments was determined. The results of this analysis revealed that the untreated control cultures healed slower than the cultures treated with bFGF-CBD-CBD (CBD-FGF) or super FGF (sFGF) (FIG. 3). In fact, wound healing was accelerated by 11% in the cultures treated with bFGF-CBD-CBD as compared to the untreated control cells. Notably, the baseline healing response that was observed in the control cultures was expected because the cells were grown in media containing 10% FBS, which contains growth factors that stimulate the recovery cultured cells. These results provide promising evidence that the bFGF-CBD-CBD fusion protein will be effective in vivo.

Second, cell viability of cardiomyocytes and cardiac fibroblasts was determined in the presence of the bFGF-CBD-CBD fusion protein using a live/dead cell viability assay [3]. The cardiac myocytes were obtained at the same time that the fibroblasts were isolated. Human cardiac fibroblasts were seeded onto collagen-coated 25 mm coverslips and placed in a 6-well plate at 200,000 cells per well. After the cells were grown to 80% confluency, they were treated with wild-type FGF (wtFGF), super FGF (sFGF), the bFGF-CBD-CBD fusion protein (FGF-CBD), or no FGF (control). After 24 hours had passed, a live/dead assay kit was used to quantify the percentage of live cells in three samples according to manufacturer's protocol. This assay was repeated after 48 hours using three additional samples. As a negative control, cell-seeded coverslips were treated with methanol for 30 minutes at the 24- and 48-hour timepoints before they were stained. The samples were imaged on a fluorescence microscope to quantify the number of live cells (FIG. 4A). All data points were normalized to the 24-hour control, and a quantitative analysis revealed that the number of cells at 48 hours was significantly increased as compared to at 24 hours in all treatment groups (FIG. 4B; * denotes $p < 0.001$). These

results demonstrate that the bFGF-CBD-CBD fusion protein does not negatively impact cell viability.

Additionally, in future work, the ability of bFGF-CBD-CBD to activate cardioprotective signaling mechanisms, namely PI3K-Akt, MEK1/2, and PPAR γ signaling, will be assessed via western blotting and ELISA. Relative expression of p-ERK1/2 and p-Akt will be assayed, and PPAR γ expression will be determined using commercially available assay kits [2].

Upon successful completion of the in vitro efficacy tests described above, the bFGF-CBD-CBD fusion protein will be tested for in vivo biocompatibility. First, the pro-fibrotic potential and biocompatibility of bFGF-CBD-CBD will be tested in a chronic rat dorsal subcutaneous implantation model. Sterile samples comprising bFGF-CBD-CBD, bFGF, or no FGF will be prepared in a carrier (e.g., collagen powder or poly(lactic-co-glycolic acid) (PLGA)). Sprague Dawley rats will be anesthetized with 3-5% isoflurane in 100% oxygen and maintained at 1-2% isoflurane using a nose cone for the entire duration of the surgery. Adequate depth of anesthesia will be checked with a toe pinch, and the rats will be positioned in a prone position on a warming pad (37° C.). Buprenorphine (0.6 mg/kg) will be administered for analgesia. The dorsal side of the rats will be shaved and treated with a depilatory to remove any traces of hair. The skin will be sterilized with an alcohol wipe followed by a povidone-iodine wipe. Using a scalpel, sterile curved surgical scissors, and tweezers, we will create four ~15 mm diameter sub-dermal pouches for injection of the drug preparations (i.e., one for each of the three preparations and sham (untreated) control) [4, 5]. The incisions will be closed with sterile 4-0 Vicryl sutures, topical tetracycline will be applied, and the wound will be covered with an occlusive dressing (i.e., Tegaderm™). The animals will then be revived from anesthesia with continuous vital sign monitoring. The rats will be closely monitored for the first 1-3 days post-surgery, and 0.3 mg/kg buprenorphine will be administered as needed. The rats will be administered analgesic immediately after surgery, 12 hours post-surgery, and whenever they appear to be in distress/pain. Rats will be sacrificed at 1, 4 and 12 weeks for analysis. The rats will be euthanized and a sterile 10 mm biopsy punch will be used to excise the scaffold and skin tissue around the wound area. This excised tissue will be divided in two. One half of the tissue will be placed in OCT and sectioned for histology, and the other half will be snap frozen for gene and protein analysis [4-7]. Histology sections will be stained with hematoxylin & eosin and Masson's trichrome to assess scar tissue/fibrosis formation and immune cell infiltration. Immunohistochemistry will be used to assess CD68 (a macrophage marker), CD163 (an activated macrophage marker), and α -SMA (an activated fibrogenic cell marker) expression as indicators of inflammation [8].

Second, the hemocompatibility of bFGF-CBD-CBD will be tested on human blood obtained from healthy adult volunteers in a standard static incubation model [9]. Blood will be sampled after 1, 6, 12, and 24 hours of contact with bFGF-CBD-CBD, bFGF, or carrier alone and assessed for platelet activation, complement activation, inflammation, and immune cell activation using commercially available assay kits.

REFERENCES FOR EXAMPLE 2

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6. Galiano, R. D., et al., *Quantitative and reproducible murine model of excisional wound healing*. Wound Repair Regen, 2004. 12(4): p. 485-92.
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8. Duijvelshoff, R., et al., *Host Response and Neo-Tissue Development during Resorption of a Fast Degrading Supramolecular Electrospun Arterial Scaffold*. Bioengineering (Basel), 2018. 5(3).
9. Weber, M., et al., *Blood-Contacting Biomaterials: In Vitro Evaluation of the Hemocompatibility*. Front Bioeng Biotechnol, 2018. 6: p. 99.

Example 3: A Second Plasmid for Expression of the bFGF-CBD-CBD Fusion Protein

The following example describes how the inventors designed and tested a second plasmid for expression of the bFGF-CBC-CBD fusion protein. This second plasmid differs from the plasmid used in Examples 1 and 2 in that (1) it contains a different promoter and regulatory element (i.e., the tac promoter and lacI^q repressor as opposed to the araBAD promoter and araC gene), and (2) it contains a different protein purification tag (i.e., a glutathione-S-transferase (GST) tag as opposed to a His-tag).

While the plasmid used in Examples 1 and 2 offers high yield for protein production solubility of the produced protein was not high. This second plasmid produced higher yield soluble protein. Further, this second plasmid was easy to isolate and contained fewer byproducts.

Construction of the expression system. Two single-stranded oligonucleotides (Collagenase site-Forward: 5'-AATTAGCGGCGGAGGTTTCAGGTCCTCTGG-GAATCGCAGGTCGGTCCGGCGGAGGTA GC (SEQ ID NO:76), and Collagenase-site-Reverse: 5'-GC-TACCTCCGCCGGACGGACCTGCGAT-TCCCAGAGGACCTGAACCTCCGCCGCT (SEQ ID NO:77)) were annealed to form a double-stranded DNA fragment encoding the cleavage site of matrix metalloproteinase-1 (MMP-1). The resulting double-stranded DNA fragment was ligated into the pCHG112-bFGF vector, which was digested with EcoRI and SmaI. Insertion of the DNA fragment was confirmed by DNA sequencing. The recombinant plasmid was named pOU693. Competent BL21-CodonPlus-RIL *E. coli* cells (Agilent Technologies Japan) were transformed with pOU693.

Protein purification. Transformed BL21-CodonPlus-RIL were grown in 2xYT medium (16 g Tryptone, 10 g Yeast extract, and 5 g NaCl per liter) supplemented with 2% (wt/vol) glucose, 50 µg/ml ampicillin, and 30 µg/ml

chloramphenicol at 37° C. with shaking at 150 rpm to an optical density of 0.5 at 600 nm. Then, the temperature was shifted down to 25° C., and the cells were grown to an optical density of 0.7 at 600 nm. Expression of the GST-tagged fusion protein was induced by the addition of 1 mM isopropyl-β-D-thiogalactopyranoside (IPTG), and the cells were cultured for 5 hours at 25° C. Phenylmethylsulfonyl fluoride (PFSF) was added to the culture to a final concentration of 1 mM, and the cells were harvested by centrifugation at 6,000 rpm for 10 min at 4° C. The cell pellet obtained from 2-liter culture was suspended in 40 ml of 50 mM Tris-HCl (pH 7.5) containing 0.5 M NaCl and 1 mM PMSF and was disrupted in a French pressure cell at 10,000 psi. The lysate was supplemented with 1% Triton X-100, and stirred for 30 min at 4° C., followed by centrifugation at 15,000 rpm for 30 min at 4° C. twice.

The resulting supernatant was mixed with 10 ml of glutathione-Sepharose (GE Healthcare) beads and stirred for 30 min at 4° C. The beads were washed five times with 10 mM Tris-HCl (pH 7.5), 0.5 M NaCl. The slurry was poured into a column. The fusion protein was eluted with 50 mM Tris-HCl (pH 8.0) containing 0.5 M NaCl and 10 mM reduced glutathione. Fractions containing significant quantities of the fusion protein were combined, and the combined fractions were treated with 10 units of thrombin per mg of eluted protein at 25° C. overnight. The cleavage products were run on an SDS-PAGE gel that was stained with Coomassie blue for protein visualization (FIG. 5). The results of these gels demonstrate that treatment with thrombin resulted in effective removal of the GST tag.

Next, the protein solution was mixed with 2 ml of heparin-Sepharose beads (GE Healthcare) and stirred for 3 hours at 4° C. The beads were washed three times with 10 mM Tris-HCl (pH 7.5), 0.5 M NaCl. The slurry was poured into an EconoColumn, and protein was eluted using a linear gradient of 0.5-2.0 M NaCl in 50 mM Tris-HCl (pH 7.5). Unbound and bound fractions were analyzed by SDS-PAGE (FIG. 5).

Fractions containing significant quantities of the fusion protein were pooled and were dialyzed against 50 mM TrisHCl (pH7.5)+0.5 M NaCl three times or against 50 mM TrisHCl (pH7.5)+0.1 M NaCl twice. SDS-PAGE analysis of the dialyzed protein indicates that the bFGF-CBD-CBD fusion protein was soluble in a physiological buffer at concentrations appropriate for purification and storage (FIG. 6). This result was surprising given that previous collagen-binding bFGF fusion proteins precipitated under these conditions. This improved solubility is attributed to the insertion of the MMP cleavage site. The protein concentration of the final pooled sample was measured using a BCA Protein Assay kit (ThermoFisher Scientific). The yield was 2.36 mg per liter of culture.

Finally, the ability of the purified bFGF-CBD-CBD fusion protein to bind to collagen was tested. 200 pmol of bFGF-CBC-CBD was dissolved in 100 µl of 50 mM TrisHCl (pH7.5), 0.1 M NaCl, 1 mM CaCl₂, and mixed with 10 mg of collagen powder prewashed in the same buffer. Two types of collagen were tested: Sigma collagen Type I (C-9879) and MP collagen (insoluble, Cat No 160083). After incubation for 30 minutes at 4° C., 15 µl each of the unbound fractions were analyzed by SDS-PAGE. The results of this analysis demonstrate that the purified bFGF-CBD-CBD fusion protein binds to collagen (FIG. 7).

SEQUENCE LISTING

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic - FGF2-MMP1 cleavage site-tandem CBD fusion protein

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Arg Leu Tyr Cys Lys Asn Gly Gly Phe Phe Leu Arg Ile His Pro Asp
35        40        45
Gly Arg Val Asp Gly Val Arg Glu Lys Ser Asp Pro His Ile Lys Leu
50        55        60
Gln Leu Gln Ala Glu Glu Arg Gly Val Val Ser Ile Lys Gly Val Cys
65        70        75        80
Ala Asn Arg Tyr Leu Ala Met Lys Glu Asp Gly Arg Leu Leu Ala Ser
85        90        95
Lys Cys Val Thr Asp Glu Cys Phe Phe Phe Glu Arg Leu Glu Ser Asn
100       105       110
Asn Tyr Asn Thr Tyr Arg Ser Arg Lys Tyr Thr Ser Trp Tyr Val Ala
115       120       125
Leu Lys Arg Thr Gly Gln Tyr Lys Leu Gly Ser Lys Thr Gly Pro Gly
130       135       140
Gln Lys Ala Ile Leu Phe Leu Pro Met Ser Ala Lys Ser Gly Ile Ser
145       150       155       160
Gly Gly Gly Ser Gly Pro Leu Gly Ile Ala Gly Pro Ser Gly Gly Gly
165       170       175
Ser Gly Thr Thr Thr Pro Ile Thr Lys Glu Met Glu Pro Asn Asp Asp
180       185       190
Ile Lys Glu Ala Asn Gly Pro Ile Val Glu Gly Val Thr Val Lys Gly
195       200       205
Asp Leu Asn Gly Ser Asp Asp Ala Asp Thr Phe Tyr Phe Asp Val Lys
210       215       220
Glu Asp Gly Asp Val Thr Ile Glu Leu Pro Tyr Ser Gly Ser Ser Asn
225       230       235       240
Phe Thr Trp Leu Val Tyr Lys Glu Gly Asp Asp Gln Asn His Ile Ala
245       250       255
Ser Gly Ile Asp Lys Asn Asn Ser Lys Val Gly Thr Phe Lys Ser Thr
260       265       270
Lys Gly Arg His Tyr Val Phe Ile Tyr Lys His Asp Ser Ala Ser Asn
275       280       285
Ile Ser Tyr Ser Leu Asn Ile Lys Gly Leu Gly Asn Glu Lys Leu Lys
290       295       300
Glu Lys Glu Asn Asn Asp Ser Ser Asp Lys Ala Thr Val Ile Pro Asn
305       310       315       320
Phe Asn Thr Thr Met Gln Gly Ser Leu Leu Gly Asp Asp Ser Arg Asp
325       330       335
Tyr Tyr Ser Phe Glu Val Lys Glu Glu Gly Glu Val Asn Ile Glu Leu
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Asp Lys Lys Asp Glu Phe Gly Val Thr Trp Thr Leu His Pro Glu Ser
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Asn Ile Asn Asp Arg Ile Thr Tyr Gly Gln Val Asp Gly Asn Lys Val
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Ser Asn Lys Val Lys Leu Arg Pro Gly Lys Tyr Tyr Leu Leu Val Tyr
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   405                               410

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<223> OTHER INFORMATION: wild-type FGF1

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Cys Ser Asn Gly Gly His Phe Leu Arg Ile Leu Pro Asp Gly Thr Val
   20                               25               30

Asp Gly Thr Arg Asp Arg Ser Asp Gln His Ile Gln Leu Gln Leu Ser
   35                               40               45

Ala Glu Ser Val Gly Glu Val Tyr Ile Lys Ser Thr Glu Thr Gly Gln
   50                               55               60

Tyr Leu Ala Met Asp Thr Asp Gly Leu Leu Tyr Gly Ser Gln Thr Pro
 65           70                               75               80

Asn Glu Glu Cys Leu Phe Leu Glu Arg Leu Glu Glu Asn His Tyr Asn
   85                               90               95

Thr Tyr Ile Ser Lys Lys His Ala Glu Lys Asn Trp Phe Val Gly Leu
 100           105               110

Lys Lys Asn Gly Ser Cys Lys Arg Gly Pro Arg Thr His Tyr Gly Gln
 115           120               125

Lys Ala Ile Leu Phe Leu Pro Leu Pro Val Ser Ser Asp
 130           135               140

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<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 3

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Cys Ser Asn Gly Gly His Phe Leu Arg Ile Leu Pro Asp Gly Thr Val
   20                               25               30

Asp Gly Thr Arg Asp Arg Ser Asp Pro His Ile Gln Leu Gln Leu Leu
   35                               40               45

Ala Glu Ser Val Gly Glu Val Tyr Ile Lys Ser Thr Glu Thr Gly Gln
   50                               55               60

Tyr Leu Ala Met Asp Thr Asp Gly Leu Leu Tyr Gly Ser Gln Thr Pro
 65           70                               75               80

Asn Glu Glu Cys Leu Phe Leu Glu Arg Leu Glu Glu Asn Ser Tyr Asn
   85                               90               95

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Thr Tyr Ile Ser Lys Lys His Ala Glu Lys Asn Trp Phe Val Gly Leu
 100 105 110

Asn Lys Asn Gly Ser Cys Lys Arg Gly Pro Glu Thr His Tyr Gly Gln
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Lys Ala Ile Leu Phe Leu Pro Leu Pro Val Ser Ser Asp
 130 135 140

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Tyr Cys Lys Asn Gly Gly Phe Phe Leu Arg Ile His Pro Asp Gly Arg
 35 40 45

Val Asp Gly Val Arg Glu Lys Ser Asp Pro His Ile Lys Leu Gln Leu
 50 55 60

Gln Ala Glu Glu Arg Gly Val Val Ser Ile Lys Gly Val Cys Ala Asn
 65 70 75 80

Arg Tyr Leu Ala Met Lys Glu Asp Gly Arg Leu Leu Ala Ser Lys Cys
 85 90 95

Val Thr Asp Glu Cys Phe Phe Phe Glu Arg Leu Glu Ser Asn Asn Tyr
 100 105 110

Asn Thr Tyr Arg Ser Arg Lys Tyr Thr Ser Trp Tyr Val Ala Leu Lys
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Arg Thr Gly Gln Tyr Lys Leu Gly Ser Lys Thr Gly Pro Gly Gln Lys
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 20 25 30

Tyr Cys Lys Asn Gly Gly Phe Phe Leu Arg Ile His Pro Asp Gly Arg
 35 40 45

Val Asp Gly Val Arg Glu Lys Ser Asp Pro His Ile Lys Leu Gln Leu
 50 55 60

Leu Ala Glu Glu Arg Gly Val Val Ser Ile Lys Gly Val Cys Ala Asn
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Arg Tyr Leu Ala Met Lys Glu Asp Gly Arg Leu Leu Ala Ser Lys Cys
 85 90 95

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Val Thr Asp Glu Cys Phe Phe Phe Glu Arg Leu Glu Ser Asn Ser Tyr
      100              105              110

Asn Thr Tyr Arg Ser Arg Lys Tyr Thr Ser Trp Tyr Val Ala Leu Asn
      115              120              125

Arg Thr Gly Gln Tyr Lys Leu Gly Ser Glu Thr Gly Pro Gly Gln Lys
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Ala Ile Leu Phe Leu Pro Met Ser Ala Lys Ser
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<223> OTHER INFORMATION: ColG s3a domain

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      20              25              30

Ala Asp Thr Phe Tyr Phe Asp Val Lys Glu Asp Gly Asp Val Thr Ile
      35              40              45

Glu Leu Pro Tyr Ser Gly Ser Ser Asn Phe Thr Trp Leu Val Tyr Lys
      50              55              60

Glu Gly Asp Asp Gln Asn His Ile Ala Ser Gly Ile Asp Lys Asn Asn
      65              70              75              80

Ser Lys Val Gly Thr Phe Lys Ser Thr Lys Gly Arg His Tyr Val Phe
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Lys

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      20              25              30

Asp Ile Phe Ala Leu Lys Val Leu Gly Asn Gly Thr Val Lys Ile Asn
      35              40              45

Val Thr Ser Glu His Asp Thr Gly Leu Asn Trp Val Val His His Glu
      50              55              60

Asp Asp Leu Asn Asn Tyr Leu Ala Tyr Pro Lys Thr Ser Gly Lys Thr
      65              70              75              80

Leu Ser Gly Glu Phe Glu Ala Thr Pro Gly Thr Tyr Tyr Leu Ser Val
      85              90              95

Tyr Asn Phe Asn Gly Glu Thr Ile Pro Tyr Lys Val Thr Ala Glu
      100              105              110

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<223> OTHER INFORMATION: ColG s3a domain

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Tyr Tyr Lys Asp Leu Val Asn Gly Ser Val Ser Ser Ser Asp Asn Lys
 20             25             30

Asp Thr Phe Tyr Phe Thr Val Thr Lys Pro Ser Asp Ile Thr Ile Thr
 35             40             45

Val Glu Lys Thr Asn Asn Asp Lys Ser Glu Phe Asn Trp Leu Leu Phe
 50             55             60

Ser Asp Glu Asp Lys Ser Asn Tyr Met Ala Phe Pro Asn Lys Glu Leu
 65             70             75             80

Gly Asn Gln Leu Ser Asn Thr Val Lys Ile Asn Lys Pro Gly Lys Tyr
 85             90             95

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Ser Ile Glu
 115

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<223> OTHER INFORMATION: ColG s3a domain from subtype A3

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 20             25             30

Asp Ile Phe Tyr Phe Asn Val Thr Lys Pro Ser Asp Ile Thr Ile Asn
 35             40             45

Val Glu Lys Ile Asn Lys Asp Lys Ser Glu Phe Ser Trp Leu Leu Phe
 50             55             60

Ser Glu Glu Asp Lys Ser Asn Tyr Ile Thr Tyr Pro Asn Lys Glu Leu
 65             70             75             80

Glu Asn Leu Phe Tyr Ser Thr Val Lys Ile Asp Lys Pro Gly Lys Tyr
 85             90             95

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 100            105            110

Asn Ile Glu
 115

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<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(113)
<223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 10

Ile Asn Glu Ser Glu Pro Asn Asn Asp Phe Glu Lys Ala Asn Gln Ile
1          5          10          15

Ala Lys Ser Asn Met Leu Val Lys Gly Thr Leu Ser Glu Glu Asp Tyr
20          25          30

Ser Asp Lys Tyr Tyr Phe Asp Val Ala Lys Lys Gly Asn Val Lys Ile
35          40          45

Thr Leu Asn Asn Leu Asn Ser Val Gly Ile Thr Trp Thr Leu Tyr Lys
50          55          60

Glu Gly Asp Leu Asn Asn Tyr Val Leu Tyr Ala Thr Gly Asn Glu Gly
65          70          75          80

Thr Val Leu Lys Gly Glu Lys Thr Leu Glu Pro Gly Arg Tyr Tyr Leu
85          90          95

Ser Val Tyr Thr Tyr Asp Asn Gln Ser Gly Ala Tyr Thr Val Asn Val
100         105         110

Lys

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<210> SEQ ID NO 11
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(113)
<223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 11

Ser Gln Glu Val Gly Asn Asp Asp Thr Phe Glu Thr Ala Asn Gly Pro
1          5          10          15

Ile Lys Ile Asn Thr Asn Tyr Ser Gly Asp Leu Ser Asp Thr Asp Asn
20          25          30

Lys Asp Tyr Tyr Tyr Phe Asn Leu Asp Asn Pro Ser Asn Ile Asn Ile
35          40          45

Thr Leu Glu Asn Leu Asp Asn Lys Gly Ile Ser Trp Gln Leu Phe His
50          55          60

Glu Ser Asp Leu Asn Asn Tyr Val Ala Tyr Pro Thr Thr Ser Gly Ala
65          70          75          80

Ile Leu Asn Gly Asp Tyr Asn Ala Thr Lys Pro Gly Lys Tyr Tyr Ile
85          90          95

Leu Val Tyr Asn His Asp Lys Ser Ile Ala Asn Tyr Asn Leu Lys Val
100         105         110

Asn

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<210> SEQ ID NO 12
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(116)
<223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 12

Thr Pro Ile Thr Lys Glu Met Glu Pro Asn Asp Asp Ile Lys Glu Ala
1          5          10          15

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Asn Gly Pro Ile Val Glu Gly Val Thr Val Lys Gly Asp Leu Asn Gly
    20                      25                      30

Ser Asp Asp Ala Asp Thr Phe Tyr Phe Asp Val Lys Glu Asp Gly Asp
    35                      40                      45

Val Thr Ile Glu Leu Pro Tyr Ser Gly Ser Ser Asn Phe Thr Trp Leu
    50                      55                      60

Val Tyr Lys Glu Gly Asp Asp Gln Asn His Ile Ala Ser Gly Ile Asp
    65                      70                      75                      80

Lys Asn Asn Ser Lys Val Gly Thr Phe Lys Ser Thr Lys Gly Arg His
    85                      90                      95

Tyr Val Phe Ile Tyr Lys His Asp Ser Ala Ser Asn Ile Ser Tyr Ser
    100                     105                     110

Leu Asn Ile Lys
    115

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<210> SEQ ID NO 13
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Clostridium sporogenes
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(117)
<223> OTHER INFORMATION: ColG s3a domain

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<400> SEQUENCE: 13

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Leu Val Ile Ser Glu Lys Glu Asp Asn Asp Ser Phe Asp Lys Ala Asn
 1                      5                      10                      15

Arg Val Cys Lys Asn Gln Ser Val Leu Ala Thr Leu Asp Thr Asn Asp
    20                      25                      30

Asn Arg Asp Thr Tyr Tyr Phe Asp Ala Leu Thr Ala Gly Thr Ile Asp
    35                      40                      45

Val Ile Met Glu Asn Thr Asp Asn Asn Ser Asn Ile Phe Asn Trp Leu
    50                      55                      60

Ala Tyr Ser Ser Asp Asn Thr Asn Asn Tyr Ile Gly Tyr Ser Thr Lys
    65                      70                      75                      80

Lys Glu Gly Asn Lys Leu Leu Gly Ser Phe Lys Val Pro Lys Pro Gly
    85                      90                      95

Arg Tyr Tyr Ile Leu Ala Tyr Lys Asn Ser Ser Asn Lys Ile Asn Tyr
    100                     105                     110

Lys Leu Thr Ile Asn
    115

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<210> SEQ ID NO 14
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Clostridium botulinum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(117)
<223> OTHER INFORMATION: ColG s3a domain from type B

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<400> SEQUENCE: 14

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Ile Asn Val Asp Glu Glu Glu Tyr Asn Asp Asp Phe Glu Cys Ala Asn
 1                      5                      10                      15

Asn Ile Phe Lys Asn Gln Ile Met Ser Gly Asn Leu Asp Ser Ser Asp
    20                      25                      30

Lys Cys Asp Thr Phe Ser Phe Asn Ala Leu Ser Ala Gly Thr Ile Asn
    35                      40                      45

Val Thr Leu Glu Asn Ser Asn Ser Asp Ser Ser Thr Val Asn Trp Leu

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50	55	60
Ala Tyr Ser Ser Glu Asp Thr Asp Asn Tyr Ile Gly Tyr Ala Ser Glu		
65	70	75 80
Asn Asp Gly Asn Lys Phe Ser Gly Lys Phe Lys Val Asn Lys Pro Gly		
	85	90 95
Lys Tyr Tyr Ile Val Ala Tyr Glu Val Asn Gly Ala Asp Ser Lys Tyr		
	100	105 110
Lys Leu Lys Val Asp		
	115	

<210> SEQ ID NO 15
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium tetani
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(117)
 <223> OTHER INFORMATION: ColG s3a domain from strain E88

<400> SEQUENCE: 15

Asn Val Ile Tyr Glu Lys Glu Asn Asn Asp Ser Phe Asp Lys Ala Asn
1 5 10 15
Lys Ile His Lys Asn Gln Ile Val Met Ala Thr Leu Asp Thr Glu Asp
20 25 30
Tyr Arg Asp Thr Phe Tyr Phe Asp Ala Leu Thr Ser Gly Ser Ile Asp
35 40 45
Ile Thr Ile Glu Asn Ile His Gly Asn Ser Asp Ala Phe Asn Trp Leu
50 55 60
Val Tyr Asn Asp Glu Asp Leu Asn Asn Tyr Ile Ala Tyr Pro Thr Lys
65 70 75 80
Lys Glu Asp Asn Lys Leu Met Gly Ser Phe Lys Val His Lys Pro Gly
85 90 95
Arg Tyr Tyr Ile Leu Val Tyr Lys Thr Ser Leu Asn Lys Val Asn Tyr
100 105 110
Lys Leu Asn Ile Ser
115

<210> SEQ ID NO 16
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium perfringens
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(115)
 <223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 16

Glu Val Ile Asn Glu Ser Glu Pro Asn Asn Asp Phe Glu Lys Ala Asn
1 5 10 15
Gln Ile Ala Lys Ser Asn Met Leu Val Lys Gly Thr Leu Ser Glu Glu
20 25 30
Asp Tyr Ser Asp Lys Tyr Tyr Phe Asp Val Ala Lys Lys Gly Asn Val
35 40 45
Lys Ile Thr Leu Asn Asn Leu Asn Ser Val Gly Ile Thr Trp Thr Leu
50 55 60
Tyr Lys Glu Gly Asp Leu Asn Asn Tyr Val Leu Tyr Ala Thr Gly Asn
65 70 75 80
Asp Gly Thr Val Leu Lys Gly Glu Lys Thr Leu Glu Pro Gly Arg Tyr
85 90 95

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Tyr Leu Ser Val Tyr Thr Tyr Asp Asn Gln Ser Gly Thr Tyr Thr Val
 100 105 110

Asn Val Lys
 115

<210> SEQ ID NO 17
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Brevibacillus brevis
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(115)
 <223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 17

Asp Gly Ser Glu Thr Glu Gln Pro Asn Pro Asn Pro Glu Glu Ser Ser
 1 5 10 15

Leu Ser Leu Gly Lys Pro Ile Thr Gly Ile Ile His Pro Gln Lys Pro
 20 25 30

Ser Gln Glu Phe Arg Leu Asp Val Lys Ser Ala Gln Gln Leu Gln Val
 35 40 45

Glu Met Glu Thr Lys Gln Gly Asp Gly Val Ala Trp Leu Val Phe His
 50 55 60

Glu Ala Asp Arg Glu Asn Tyr Ile Ser Tyr Pro Thr Lys Arg Glu Gly
 65 70 75 80

Asn Lys Leu Ile Gly Ser Phe Asp Ala Lys Pro Gly Thr Tyr Tyr Val
 85 90 95

Thr Ala Tyr Thr Tyr Arg Thr Glu Gln Glu Asp Gln Pro Phe Arg Leu
 100 105 110

Leu Val Thr
 115

<210> SEQ ID NO 18
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Paenibacillus dendritiformis
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(114)
 <223> OTHER INFORMATION: ColG s3a domain

<400> SEQUENCE: 18

Pro Asn Ala Asp His Glu Pro Asn Asp Ser Trp Glu Gln Ala Val Pro
 1 5 10 15

Leu Asp Gly Thr Gly Val Pro Val Ser Gly Lys Leu Ser Asp Thr Asp
 20 25 30

Arg Val Asp Val Tyr Arg Phe Asp Ala Gly Lys Ala Glu Gln Trp Thr
 35 40 45

Ile Glu Leu Glu Thr Glu Gln Ala Gln Ser Val Ala Trp Val Val His
 50 55 60

His Glu Ser Asp Leu Asn Asn Tyr Ala Ala Tyr Pro Thr Gln Val Glu
 65 70 75 80

Gly Thr Ser Val Ala Gly Ser Val Asp Ala Val Pro Gly Thr Tyr Tyr
 85 90 95

Val Tyr Val Tyr Ser Val Gly Asn Gly Glu Gln Ser Tyr Arg Leu Val
 100 105 110

Val Gln

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<210> SEQ ID NO 19
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(116)
<223> OTHER INFORMATION: ColG s3a domain

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<400> SEQUENCE: 19

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Pro Lys Gly Ser Gln Glu Val Gly Asn Asp Asp Thr Phe Glu Thr Ala
1          5          10          15
Asn Gly Pro Ile Lys Ile Asn Thr Asn Tyr Ser Gly Asp Leu Ser Asp
20        25        30
Thr Asp Asn Lys Asp Tyr Tyr Tyr Phe Asn Leu Asp Asn Pro Ser Asn
35        40        45
Ile Asn Ile Thr Leu Glu Asn Leu Asp Asn Lys Gly Ile Ser Trp Gln
50        55        60
Leu Phe His Glu Ser Asp Leu Asn Asn Tyr Val Ala Tyr Pro Thr Thr
65        70        75        80
Ser Gly Ala Ile Leu Asn Gly Asp Tyr Asn Ala Thr Lys Pro Gly Lys
85        90        95
Tyr Tyr Ile Leu Val Tyr Asn His Asp Lys Ser Ile Ala Asn Tyr Asn
100       105       110
Leu Lys Val Asn
115

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<210> SEQ ID NO 20
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 20

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Leu Lys Glu Lys Glu Asn Asn Asp Ser Ser Asp Lys Ala Thr Val Ile
1          5          10          15
Pro Asn Phe Asn Thr Thr Met Gln Gly Ser Leu Leu Gly Asp Asp Ser
20        25        30
Arg Asp Tyr Tyr Ser Phe Glu Val Lys Glu Glu Gly Glu Val Asn Ile
35        40        45
Glu Leu Asp Lys Lys Asp Glu Phe Gly Val Thr Trp Thr Leu His Pro
50        55        60
Glu Ser Asn Ile Asn Asp Arg Ile Thr Tyr Gly Gln Val Asp Gly Asn
65        70        75        80
Lys Val Ser Asn Lys Val Lys Leu Arg Pro Gly Lys Tyr Tyr Leu Leu
85        90        95
Val Tyr Lys Tyr Ser Gly Ser Gly Asn Tyr Glu Leu Arg Val Asn
100       105       110

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<210> SEQ ID NO 21
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Bacillus sp.
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 21

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Lys Thr Glu Ile Glu Pro Asn Asn Arg Pro Glu Glu Ala Thr Met Leu
 1 5 10 15
 Pro Phe Asn Thr Pro Leu Ser Gly Ser Leu Met Glu Asp Asp His Thr
 20 25 30
 Asp Val Tyr Glu Phe Asn Val Thr Ser Pro Lys Glu Ile Asp Ile Ser
 35 40 45
 Val Leu Asn Glu Asn Gln Ile Gly Met Thr Trp Val Leu Tyr His Glu
 50 55 60
 Ser Asp Ser Gln Asn Tyr Ala Ser Phe Gly Gln Glu Asp Gly Asn Met
 65 70 75 80
 Ile Asn Gly Lys Trp Asn Ala Lys Pro Gly Lys Tyr Tyr Leu Tyr Val
 85 90 95
 Tyr Lys Phe Glu Asn Glu Asn Gly Thr Tyr Thr Val His Val Gln
 100 105 110

<210> SEQ ID NO 22
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Lysinibacillus sphaericus
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(111)
 <223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 22

Lys Ala Glu Ile Glu Pro Asn Asn Arg Pro Glu Glu Ala Thr Ile Leu
 1 5 10 15
 Pro Phe Asn Thr Pro Leu Lys Gly Arg Leu Met Asp Asp Asp His Thr
 20 25 30
 Asp Val Tyr Glu Phe Asn Val Thr Ser Pro Lys Glu Leu Asp Ile Ser
 35 40 45
 Val Leu Asn Glu Asn Arg Ile Gly Met Thr Trp Val Leu Tyr His Glu
 50 55 60
 Ser Asp Ser Gln Asn Tyr Ala Ser Phe Gly Gln Glu Glu Gly Asn Met
 65 70 75 80
 Ile Asn Gly Lys Leu His Ala Glu Pro Gly Lys Tyr Tyr Leu Tyr Val
 85 90 95
 Tyr Lys Phe Glu Asn Glu Asn Gly Thr Tyr Thr Val Gln Val Gln
 100 105 110

<210> SEQ ID NO 23
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Brevibacillus brevis
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(111)
 <223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 23

Pro Thr Glu Val Glu Pro Asn Asn Ser Phe Asp Asp Ala Asn Thr Leu
 1 5 10 15
 Gln Leu Gly Lys Glu Ile Ser Gly Gln Thr Asp Arg Thr Asp Asp Lys
 20 25 30
 Asp Thr Tyr Met Ile Gln Val Glu Glu Glu Gly Val Ile Gln Val Thr
 35 40 45
 Val Ser Ser Glu Lys Asp Glu Gly Leu Asn Trp Val Val Phe His Glu
 50 55 60

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Asp Asp Leu Lys Thr Tyr Phe Ala Tyr Pro Lys Thr Thr Gly Lys Lys
65 70 75 80

Leu Thr Gly Glu Phe Glu Ala Lys Pro Gly Lys Tyr Tyr Leu Leu Val
85 90 95

Tyr Asn Thr Asn Asn Thr Lys Ile Pro Tyr Lys Ala Ile Val Asn
100 105 110

<210> SEQ ID NO 24
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Clostridium perfringens
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1) .. (112)
<223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 24

Ile Lys Glu Val Glu Asn Asn Asn Asp Phe Asp Lys Ala Met Lys Val
1 5 10 15

Asp Ser Asn Ser Lys Ile Val Gly Thr Leu Ser Asn Asp Asp Leu Lys
20 25 30

Asp Ile Tyr Ser Ile Asp Ile Lys Asn Pro Ser Asp Leu Asn Ile Val
35 40 45

Val Glu Asn Leu Asp Asn Ile Lys Met Asn Trp Leu Leu Tyr Ser Ala
50 55 60

Asp Asp Leu Ser Asn Tyr Val Asp Tyr Ala Asn Ala Asp Gly Asn Lys
65 70 75 80

Leu Ser Asn Thr Cys Lys Leu Asn Pro Gly Lys Tyr Tyr Leu Cys Val
85 90 95

Tyr Gln Phe Glu Asn Ser Gly Thr Gly Asn Tyr Thr Val Asn Leu Gln
100 105 110

<210> SEQ ID NO 25
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Clostridium sporogenes
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1) .. (115)
<223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 25

Ile Ser Glu Lys Glu Asp Asn Asp Ser Phe Asp Lys Ala Asn Arg Val
1 5 10 15

Gly Lys Asn Gln Thr Val Leu Ala Thr Leu Asp Thr Lys Asp Asn Arg
20 25 30

Asp Thr Tyr Tyr Phe Asp Ala Leu Ala Ala Arg Thr Ile Asp Ile Val
35 40 45

Met Glu Asn Thr Asp Asn Asn Ser Thr Ile Phe Asn Trp Leu Ala Tyr
50 55 60

Ser Ser Asp Asn Thr Asn Asn Tyr Ile Gly Tyr Pro Thr Lys Lys Glu
65 70 75 80

Gly Asn Lys Leu Met Gly Ser Phe Lys Val Pro Lys Pro Gly Arg Tyr
85 90 95

Tyr Ile Leu Ala Tyr Lys Asn Ser Ser Asn Lys Ile Asn Tyr Lys Leu
100 105 110

Thr Ile Asn
115

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<210> SEQ ID NO 26
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Clostridium botulinum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(115)
<223> OTHER INFORMATION: ColG s3b domain from strain A3

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<400> SEQUENCE: 26
Ile Ser Glu Lys Glu Asp Asn Asn Ser Phe Asp Lys Ala Asn Arg Val
1      5      10      15
Cys Lys Asn Gln Ser Val Ile Ala Thr Leu Asp Thr Asn Asp Pro Arg
20     25     30
Asp Thr Tyr Tyr Phe Asp Ala Leu Thr Ala Gly Asn Ile Glu Val Thr
35     40     45
Met Gly Asn Thr Asp Asn Ser Ser Asn Glu Phe Asn Trp Leu Ala Tyr
50     55     60
Ser Ser Asp Asn Thr Asn Asn Tyr Ile Gly Tyr Ala Thr Lys Arg Glu
65     70     75     80
Gly Asn Lys Ile Thr Gly Asn Phe Lys Val Asp Lys Pro Gly Arg Tyr
85     90     95
Tyr Ile Val Ala Tyr Lys Thr Ser Ser Asn Lys Ile Asn Tyr Lys Leu
100    105    110
Asn Ile Lys
115

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<210> SEQ ID NO 27
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(113)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 27
Gly Val Glu Gln Glu Asp Asn Asn Ser Phe Glu Lys Ala Asn Pro Phe
1      5      10      15
Ser Ile Asn Gln Leu Val Lys Gly Glu Leu Asp Asn Asn Lys Asp Thr
20     25     30
Ser Asp Tyr Phe Lys Phe Glu Val Lys Glu Asp Ala Gln Leu Asn Ile
35     40     45
Ser Leu Glu Lys Thr Glu Gly Asp Gly Val Asn Trp Leu Leu Phe Lys
50     55     60
Asp Ser Asp Leu Glu Asn Tyr Ile Ala Ser Pro Thr Glu Ser Ile Asp
65     70     75     80
Asn Lys Leu Asn Gly Lys Val Asp Leu Lys Val Gly Thr Tyr Tyr Leu
85     90     95
Glu Val Tyr Gly Tyr Gly Ser Ser Pro Val Lys Tyr Asn Phe Lys Val
100    105    110
Thr

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<210> SEQ ID NO 28
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(112)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 28

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Leu Lys Glu Lys Glu Asn Asn Asp Ser Ser Asp Lys Ala Thr Val Ile
1          5          10          15
Pro Asn Phe Asn Thr Thr Met Gln Gly Ser Leu Leu Gly Asp Asp Ser
          20          25          30
Arg Asp Tyr Tyr Ser Phe Glu Val Lys Glu Glu Gly Glu Val Asn Ile
          35          40          45
Glu Leu Asp Lys Lys Asp Glu Phe Gly Val Thr Trp Thr Leu His Pro
          50          55          60
Glu Ser Asn Ile Asn Asp Arg Ile Thr Tyr Gly Gln Val Asp Gly Asn
65          70          75          80
Lys Val Ser Asn Lys Val Lys Leu Arg Pro Gly Lys Tyr Tyr Leu Leu
          85          90          95
Val Tyr Lys Tyr Ser Gly Ser Gly Asn Tyr Glu Leu Arg Val Asn Lys
          100          105          110

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<210> SEQ ID NO 29

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Clostridium sporogenes

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (1)..(111)

<223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 29

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Ile His Glu Lys Glu Asn Asn Asp Ser Phe Glu Ser Ala Asn Lys Ile
1          5          10          15
Val Pro Asp Ala Pro Val Leu Gly Ser Leu Asn Gly Glu Asp Leu Arg
          20          25          30
Asp Ile Tyr Ser Phe Asp Ile Lys Glu Thr Lys Asp Leu Asn Ile Lys
          35          40          45
Leu Thr Asn Leu Asn Asn Leu Gly Leu Thr Trp Thr Leu Tyr Lys Glu
          50          55          60
Ser Asp Leu Thr Asn Tyr Ile Ala Tyr Gly Ser Lys Leu Gly Asn Thr
65          70          75          80
Ile Val Gly Asn Cys His Val Thr Pro Gly Lys Tyr Tyr Leu Tyr Val
          85          90          95
Tyr Lys Tyr Ser Gly Asn Ser Gly Asn Tyr Ser Leu Ile Ile Lys
          100          105          110

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<210> SEQ ID NO 30

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Clostridium botulinum

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (1)..(111)

<223> OTHER INFORMATION: ColG s3b domain from type B

<400> SEQUENCE: 30

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Ile Lys Glu Glu Ile Asn Asp Asp Ser Phe Asp Ser Ala Thr Lys Ile
1          5          10          15
Lys Ala Asn Ser Thr Ile Thr Asp Thr Leu Asn Gly Glu Asp Asn Lys
          20          25          30
Asp Ile Cys Tyr Phe Asn Val Asn Asn Asn Ser Asp Leu Asn Ile Glu
          35          40          45
Leu Asn Ser Leu Thr Asn Leu Gly Val Ala Trp Gln Leu Phe Ser Glu

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50	55	60
Glu Asp Leu Asp Asn Tyr Ile Ala Tyr Gly Ser Gln Ser Gly Asp Ser		
65	70	75 80
Ile Val Gly Thr Ala Asn Val Gln Pro Gly Lys Tyr Tyr Leu Leu Ile		
	85	90 95
Tyr Lys Tyr Thr Gln Ala Asp Gly Ser Tyr Thr Phe Thr Ile Lys		
	100 105	110

<210> SEQ ID NO 31
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium tetani
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(111)
 <223> OTHER INFORMATION: ColG s3b domain from strain E88

<400> SEQUENCE: 31

Ile His Glu Lys Glu Asn Asn Asp Ser Phe Glu Thr Ala Asn Lys Ile		
1	5	10 15
Thr Leu Asp Thr Leu Val Leu Gly Asn Leu Asp Tyr Lys Asp Val Ser		
	20	25 30
Asp Ile Tyr Ser Phe Asp Ile Glu Asn Thr Lys Asp Leu Asn Ile Lys		
	35	40 45
Leu Thr Asn Leu Asn Asn Leu Gly Ile Ala Trp Asn Leu Tyr Lys Glu		
	50	55 60
Ser Asp Leu Asn Asn Tyr Ile Ala Tyr Gly Ala Lys Ser Asp Asn Ala		
65	70	75 80
Ile Val Gly Lys Cys Asn Leu Ser Pro Gly Lys Tyr Tyr Leu Tyr Val		
	85	90 95
Tyr Lys Tyr Ser Gly Asp Lys Gly Asn Tyr Ser Val Ile Ile Asn		
	100 105	110

<210> SEQ ID NO 32
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium perfringens
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(114)
 <223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 32

Ile Lys Glu Val Glu Asn Asn Asn Asp Phe Asp Lys Ala Met Lys Val		
1	5	10 15
Asp Ser Asn Ser Lys Ile Val Gly Thr Leu Ser Asn Asp Asp Leu Lys		
	20	25 30
Asp Ile Tyr Ser Ile Asp Ile Gln Asn Pro Ser Asp Leu Asn Ile Val		
	35	40 45
Val Glu Asn Leu Asp Asn Ile Lys Met Asn Trp Leu Leu Tyr Ser Ala		
	50	55 60
Asp Asp Leu Ser Asn Tyr Val Asp Tyr Ala Asn Ala Asp Gly Asn Lys		
65	70	75 80
Leu Ser Asn Thr Cys Lys Leu Asn Pro Gly Lys Tyr Tyr Leu Cys Val		
	85	90 95
Tyr Gln Phe Glu Asn Ser Gly Thr Gly Asn Tyr Ile Val Asn Leu Gln		
	100 105	110

Asn Lys

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<210> SEQ ID NO 33
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Brevibacillus brevis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(115)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 33

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Tyr Gln Glu Asn Glu Ser Asn Asp Ser Thr Glu Gln Ala Asn Gly Pro
1          5          10          15

Leu Lys Ile Gly Thr Thr Val Ser Gly Asp Met Lys Gly Asn Asp Trp
          20          25          30

Gln Asp Ile Phe Ala Phe Gln Val Asp Lys Pro Glu Glu Ile Arg Ile
          35          40          45

Ser Leu Asn Pro Gln Glu Gly Gln Gly Val Thr Trp Met Leu Phe His
          50          55          60

Glu Gly Asn Leu Asp Gln Pro Val Thr Tyr Pro Gln Glu Arg Glu Gly
          65          70          75          80

Asn Leu Gln Ser Ala His Tyr Gln Val Lys Pro Gly Arg Tyr Phe Leu
          85          90          95

Tyr Val Tyr Lys Tyr Gln Asn Glu Asp Ile Val Tyr Thr Val Glu Thr
          100          105          110

Lys Gln Arg
          115

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<210> SEQ ID NO 34
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Paenibacillus dendritiformis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(114)
<223> OTHER INFORMATION: ColG s3b domain

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<400> SEQUENCE: 34

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Phe Glu Glu Thr Glu Pro Asn Asp Thr Pro Glu Thr Ala Asn Gly Pro
1          5          10          15

Ile Pro Ala Gly Arg Pro Val Val Gly Thr Leu Asn Gly Ser Asp Lys
          20          25          30

Gln Asp Val Phe Ile Ile Asp Val Asp Gln Pro Ala Glu Leu Gln Ile
          35          40          45

Glu Leu Glu Arg Arg Leu Gly Ser Gly Val Asn Trp Ile Leu Tyr Arg
          50          55          60

Glu Gly Asp Thr Asp Arg Pro Leu Leu Tyr Pro Ser Glu Val Glu Gly
          65          70          75          80

Asn Arg Met Ser Gly Gly Phe Ala Ala Glu Ala Gly Arg Tyr His Leu
          85          90          95

Tyr Val Tyr Lys Tyr Thr Asp Glu Asp Ile His Tyr Thr Leu Gln Val
          100          105          110

Gln His

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<210> SEQ ID NO 35
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(115)

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<223> OTHER INFORMATION: ColG s3b domain

<400> SEQUENCE: 35

Gly	Val	Glu	Gln	Glu	Asp	Asn	Asn	Ser	Phe	Glu	Lys	Ala	Asn	Pro	Phe
1				5					10					15	

Ser	Ile	Asn	Gln	Leu	Val	Lys	Gly	Glu	Leu	Asp	Asn	Asn	Lys	Asp	Thr
			20					25					30		

Ser	Asp	Tyr	Phe	Lys	Phe	Glu	Val	Lys	Glu	Asp	Ala	Gln	Leu	Asn	Ile
		35						40				45			

Ser	Leu	Glu	Lys	Thr	Glu	Gly	Asp	Gly	Val	Asn	Trp	Leu	Leu	Phe	Lys
	50					55					60				

Asp	Ser	Asp	Leu	Glu	Asn	Tyr	Ile	Ala	Ser	Pro	Thr	Glu	Ser	Ile	Asp
65					70					75				80	

Asn	Lys	Leu	Asn	Gly	Lys	Val	Asp	Leu	Lys	Val	Gly	Thr	Tyr	Tyr	Leu
			85						90					95	

Glu	Val	Tyr	Gly	Tyr	Gly	Ser	Ser	Pro	Val	Lys	Tyr	Asn	Phe	Lys	Val
			100					105						110	

Thr	Pro	Asn
		115

<210> SEQ ID NO 36

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Clostridium sporogenes

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (1)..(111)

<223> OTHER INFORMATION: ColG s3c domain

<400> SEQUENCE: 36

Ile	His	Glu	Lys	Glu	Asn	Asn	Asp	Ser	Phe	Glu	Ser	Ala	Asn	Lys	Ile
1				5					10					15	

Val	Leu	Asn	Ala	Pro	Ile	Leu	Gly	Ser	Leu	Asn	Gly	Glu	Asp	Leu	Arg
			20					25					30		

Asp	Ile	Tyr	Ser	Phe	Glu	Ile	Lys	Glu	Thr	Lys	Asp	Leu	Asn	Ile	Lys
		35					40					45			

Leu	Thr	Asn	Leu	Asn	Asn	Leu	Gly	Leu	Thr	Trp	Thr	Leu	Tyr	Lys	Glu
	50					55				60					

Ser	Asp	Leu	Asn	Asn	Tyr	Ile	Ala	Tyr	Gly	Ser	Lys	Leu	Gly	Ser	Thr
65					70					75				80	

Ile	Val	Gly	Asn	Cys	His	Val	Thr	Pro	Gly	Lys	Tyr	Tyr	Leu	Tyr	Val
			85						90					95	

Tyr	Lys	Tyr	Ser	Gly	Asn	Asn	Gly	Asn	Tyr	Ser	Leu	Ile	Ile	Lys
			100					105					110	

<210> SEQ ID NO 37

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Clostridium botulinum

<220> FEATURE:

<221> NAME/KEY: MISC_FEATURE

<222> LOCATION: (1)..(111)

<223> OTHER INFORMATION: ColG s3c domain from strain A3

<400> SEQUENCE: 37

Ile	Tyr	Glu	Lys	Glu	Asn	Asn	Asp	Ser	Phe	Glu	Thr	Ala	Asn	Lys	Ile
1				5					10					15	

Met	Leu	Asn	Thr	Thr	Val	Leu	Gly	Asn	Leu	Asn	Gly	Lys	Asp	Val	Arg
			20					25					30		

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Asp Ile Tyr Ser Phe Asp Ile Lys Glu Ala Lys Asp Leu Asp Ile Lys
   35               40               45

Leu Asn Asn Leu Asn Asn Leu Gly Leu Ala Trp Asn Leu Tyr Lys Glu
   50               55               60

Ser Asp Leu Asn Asn Tyr Ile Ala Tyr Gly Ser Val Ser Gly Asn Thr
   65               70               75               80

Ile Lys Gly Lys Cys Asn Val Ala Pro Gly Lys Tyr Tyr Leu Tyr Val
               85               90               95

Tyr Lys Tyr Ser Gly Asp Asn Gly Asn Tyr Ser Leu Ala Ile Lys
   100               105               110

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<210> SEQ ID NO 38
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Bacillus cereus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: AH603 domain

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<400> SEQUENCE: 38

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Leu Thr Glu Ser Glu Pro Asn Asn Arg Pro Glu Glu Ala Asn Arg Ile
 1               5               10               15

Gly Leu Asn Thr Thr Ile Lys Gly Ser Leu Ile Gly Gly Asp His Thr
   20               25               30

Asp Val Tyr Thr Phe Asn Val Ala Ser Ala Lys Asn Ile Asn Ile Ser
   35               40               45

Val Leu Asn Glu Tyr Gly Ile Gly Met Thr Trp Val Leu His His Glu
   50               55               60

Ser Asp Met Gln Asn Tyr Ala Ala Tyr Gly Gln Val Asn Gly Asn His
   65               70               75               80

Ile Glu Ala Asn Phe Asn Ala Lys Pro Gly Lys Tyr Tyr Leu Tyr Val
   85               90               95

Tyr Lys Tyr Asp Asn Gly Asp Gly Thr Tyr Glu Leu Ser Val Lys
   100               105               110

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<210> SEQ ID NO 39
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Bacillus anthracis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: ColG domain

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<400> SEQUENCE: 39

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Leu Thr Glu Ser Glu Pro Asn Asn Arg Pro Glu Glu Ala Asn Arg Ile
 1               5               10               15

Gly Leu Asn Thr Thr Ile Lys Gly Ser Leu Ile Gly Gly Asp His Thr
   20               25               30

Asp Val Tyr Thr Phe Asn Val Ala Ser Ala Lys Asn Ile Asp Ile Ser
   35               40               45

Val Leu Asn Glu Tyr Gly Ile Gly Met Thr Trp Val Leu His His Glu
   50               55               60

Ser Asp Met Gln Asn Tyr Ala Ala Tyr Gly Gln Ala Asn Gly Asn His
   65               70               75               80

Ile Glu Ala Asn Phe Asn Ala Lys Pro Gly Lys Tyr Tyr Leu Tyr Val
   85               90               95

Tyr Lys Tyr Asp Asn Gly Asp Gly Thr Tyr Glu Leu Ser Val Lys

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100              105              110

<210> SEQ ID NO 40
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Bacillus cereus
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: G9842 domain

<400> SEQUENCE: 40

Val Thr Glu Asn Glu Pro Asn Asn Glu Pro Arg Gln Ala Asn Lys Val
1          5              10              15
Asn Phe His Thr Pro Val Lys Gly Thr Leu His Asn Ser Asp Arg Val
20          25              30
Asp Val Phe Thr Phe Gln Ile Asp Ser Pro Glu Asn Ile Asn Ile Ser
35          40              45
Leu Leu Asn Glu Gln Asn Ile Gly Met Thr Trp Val Leu His His Glu
50          55              60
Ser Asp Leu Asn Asn Tyr Val Ala Tyr Gly Glu Asn Glu Gly Asn Val
65          70              75              80
Val Lys Gly Thr Tyr Asn Ala Lys Pro Gly Lys Tyr Tyr Leu Tyr Val
85          90              95
Tyr Lys Tyr Glu Asn Lys Asp Gly Ser Tyr Val Leu Asn Ile Lys
100         105         110

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<210> SEQ ID NO 41
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Bacillus mycoides
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(111)
<223> OTHER INFORMATION: ColG domain

<400> SEQUENCE: 41

Ser Val Glu Lys Glu Pro Asn Asn Ser Phe Gln Thr Ala Asn Lys Leu
1          5              10              15
Gln Leu Asn Gln Leu Leu Arg Ala Ser Leu Gly Asn Gly Asp Thr Ser
20          25              30
Asp Tyr Phe Glu Ile Asn Val Glu Thr Ala Arg Asn Leu Gln Ile Asn
35          40              45
Val Thr Lys Glu Asn Asn Ile Gly Val Asn Trp Val Leu Tyr Ser Ala
50          55              60
Ala Asp Leu Asn Asn Tyr Val Thr Tyr Ala Gln Thr Gln Gly Asn Lys
65          70              75              80
Leu Val Gly Ser Tyr Asn Ala His Pro Gly Lys Tyr Tyr Leu His Val
85          90              95
Tyr Gln Tyr Gly Gly Gly Thr Gly Asn Tyr Thr Val Glu Val Lys
100         105         110

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<210> SEQ ID NO 42
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Bacillus weihensteph
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(112)
<223> OTHER INFORMATION: ColG domain

<400> SEQUENCE: 42

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Ala Val Glu Lys Glu Pro Asn Asn Ser Phe Asp Ala Ala Asn Pro Leu
 1 5 10 15
 Ser Leu Asn Ala Leu Leu Arg Gly Asn Leu Ser Asp Gln Asp Gln Val
 20 25 30
 Asp Arg Phe Val Ile Asp Val Lys Asp Pro Lys Asp Leu Gln Ile Thr
 35 40 45
 Val Thr Asn Glu Gln Asn Leu Gly Leu Asn Trp Val Leu Tyr Ser Glu
 50 55 60
 Ser Asp Leu Asn Asn Tyr Val Thr Tyr Ala Thr Lys Arg Asp Gly Asn
 65 70 75 80
 Lys Leu Leu Gly Asn Tyr Asn Ala Lys Pro Gly Lys Tyr Tyr Leu Ser
 85 90 95
 Val Tyr Lys Tyr Gly Gly Gly Thr Gly Asn Phe Thr Val Glu Val Lys
 100 105 110

<210> SEQ ID NO 43
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium histolyticum
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(125)
 <223> OTHER INFORMATION: ColH s3 domain

<400> SEQUENCE: 43

Gly Ile Asn Ser Pro Val Tyr Pro Ile Gly Thr Glu Lys Glu Pro Asn
 1 5 10 15
 Asn Ser Lys Glu Thr Ala Ser Gly Pro Ile Val Pro Gly Ile Pro Val
 20 25 30
 Ser Gly Thr Ile Glu Asn Thr Ser Asp Gln Asp Tyr Phe Tyr Phe Asp
 35 40 45
 Val Ile Thr Pro Gly Glu Val Lys Ile Asp Ile Asn Lys Leu Gly Tyr
 50 55 60
 Gly Gly Ala Thr Trp Val Val Tyr Asp Glu Asn Asn Asn Ala Val Ser
 65 70 75 80
 Tyr Ala Thr Asp Asp Gly Gln Asn Leu Ser Gly Lys Phe Lys Ala Asp
 85 90 95
 Lys Pro Gly Arg Tyr Tyr Ile His Leu Tyr Met Phe Asn Gly Ser Tyr
 100 105 110
 Met Pro Tyr Arg Ile Asn Ile Glu Gly Ser Val Gly Arg
 115 120 125

<210> SEQ ID NO 44
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium histolyticum
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(111)
 <223> OTHER INFORMATION: ColH s3 domain

<400> SEQUENCE: 44

Gly Thr Glu Lys Glu Pro Asn Asn Ser Lys Glu Thr Ala Ser Gly Pro
 1 5 10 15
 Ile Val Pro Gly Ile Pro Val Ser Gly Thr Ile Glu Asn Thr Ser Asp
 20 25 30
 Gln Asp Tyr Phe Tyr Phe Asp Val Ile Thr Pro Gly Glu Val Lys Ile
 35 40 45

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Asp Ile Asn Lys Leu Gly Tyr Gly Gly Ala Thr Trp Val Val Tyr Asp
 50                      55                      60

Glu Asn Asn Asn Ala Val Ser Tyr Ala Thr Asp Asp Gly Gln Asn Leu
65                      70                      75                      80

Ser Gly Lys Phe Lys Ala Asp Lys Pro Gly Arg Tyr Tyr Ile His Leu
                        85                      90                      95

Tyr Met Phe Asn Gly Ser Tyr Met Pro Tyr Arg Ile Asn Ile Glu
      100                      105                      110

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<210> SEQ ID NO 45
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(121)
<223> OTHER INFORMATION: ColH domain

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<400> SEQUENCE: 45

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Pro Val Tyr Pro Ile Gly Thr Glu Lys Glu Pro Asn Asn Ser Lys Glu
 1                      5                      10                      15

Thr Ala Ser Gly Pro Ile Val Pro Gly Ile Pro Val Ser Gly Thr Ile
                20                      25                      30

Glu Asn Thr Ser Asp Gln Asp Tyr Phe Tyr Phe Asp Val Ile Thr Pro
      35                      40                      45

Gly Glu Val Lys Ile Asp Ile Asn Lys Leu Gly Tyr Gly Gly Ala Thr
      50                      55                      60

Trp Val Val Tyr Asp Glu Asn Asn Asn Ala Val Ser Tyr Ala Thr Asp
65                      70                      75                      80

Asp Gly Gln Asn Leu Ser Gly Lys Phe Lys Ala Asp Lys Pro Gly Arg
      85                      90                      95

Tyr Tyr Ile His Leu Tyr Met Phe Asn Gly Ser Tyr Met Pro Tyr Arg
      100                      105                      110

Ile Asn Ile Glu Gly Ser Val Gly Arg
      115                      120

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<210> SEQ ID NO 46
<211> LENGTH: 234
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(234)
<223> OTHER INFORMATION: ColG s3a + s3b domains

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<400> SEQUENCE: 46

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Thr Pro Ile Thr Lys Glu Met Glu Pro Asn Asp Asp Ile Lys Glu Ala
 1                      5                      10                      15

Asn Gly Pro Ile Val Glu Gly Val Thr Val Lys Gly Asp Leu Asn Gly
      20                      25                      30

Ser Asp Asp Ala Asp Thr Phe Tyr Phe Asp Val Lys Glu Asp Gly Asp
      35                      40                      45

Val Thr Ile Glu Leu Pro Tyr Ser Gly Ser Ser Asn Phe Thr Trp Leu
      50                      55                      60

Val Tyr Lys Glu Gly Asp Asp Gln Asn His Ile Ala Ser Gly Ile Asp
65                      70                      75                      80

Lys Asn Asn Ser Lys Val Gly Thr Phe Lys Ser Thr Lys Gly Arg His
      85                      90                      95

Tyr Val Phe Ile Tyr Lys His Asp Ser Ala Ser Asn Ile Ser Tyr Ser

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-continued

100	105	110
Leu Asn Ile Lys Gly Leu Gly	Asn Glu Lys Leu Lys	Glu Lys Glu Asn
115	120	125
Asn Asp Ser Ser Asp Lys Ala Thr Val Ile Pro Asn Phe Asn Thr Thr		
130	135	140
Met Gln Gly Ser Leu Leu Gly Asp Asp Ser Arg Asp Tyr Tyr Ser Phe		
145	150	155
Glu Val Lys Glu Glu Gly Glu Val Asn Ile Glu Leu Asp Lys Lys Asp		
165	170	175
Glu Phe Gly Val Thr Trp Thr Leu His Pro Glu Ser Asn Ile Asn Asp		
180	185	190
Arg Ile Thr Tyr Gly Gln Val Asp Gly Asn Lys Val Ser Asn Lys Val		
195	200	205
Lys Leu Arg Pro Gly Lys Tyr Tyr Leu Leu Val Tyr Lys Tyr Ser Gly		
210	215	220
Ser Gly Asn Tyr Glu Leu Arg Val Asn Lys		
225	230	

<210> SEQ ID NO 47
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium sporogenes
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1) .. (239)
 <223> OTHER INFORMATION: ColG s3a + s3b domains

<400> SEQUENCE: 47

Leu Val Ile Ser Glu Lys Glu Asp Asn Asp Ser Phe Asp Lys Ala Asn
1 5 10 15
Arg Val Cys Lys Asn Gln Ser Val Leu Ala Thr Leu Asp Thr Asn Asp
20 25 30
Asn Arg Asp Thr Tyr Tyr Phe Asp Ala Leu Thr Ala Gly Thr Ile Asp
35 40 45
Val Ile Met Glu Asn Thr Asp Asn Asn Ser Asn Ile Phe Asn Trp Leu
50 55 60
Ala Tyr Ser Ser Asp Asn Thr Asn Asn Tyr Ile Gly Tyr Ser Thr Lys
65 70 75 80
Lys Glu Gly Asn Lys Leu Leu Gly Ser Phe Lys Val Pro Lys Pro Gly
85 90 95
Arg Tyr Tyr Ile Leu Ala Tyr Lys Asn Ser Ser Asn Lys Ile Asn Tyr
100 105 110
Lys Leu Thr Ile Asn Gly Asp Ile Asp Lys Ala Pro Leu Lys Asn Glu
115 120 125
Ile His Glu Lys Glu Asn Asn Asp Ser Phe Glu Ser Ala Asn Lys Ile
130 135 140
Val Pro Asp Ala Pro Val Leu Gly Ser Leu Asn Gly Glu Asp Leu Arg
145 150 155 160
Asp Ile Tyr Ser Phe Asp Ile Lys Glu Thr Lys Asp Leu Asn Ile Lys
165 170 175
Leu Thr Asn Leu Asn Asn Leu Gly Leu Thr Trp Thr Leu Tyr Lys Glu
180 185 190
Ser Asp Leu Thr Asn Tyr Ile Ala Tyr Gly Ser Lys Leu Gly Asn Thr
195 200 205
Ile Val Gly Asn Cys His Val Thr Pro Gly Lys Tyr Tyr Leu Tyr Val
210 215 220

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Tyr Lys Tyr Ser Gly Asn Ser Gly Asn Tyr Ser Leu Ile Ile Lys
 225 230 235

<210> SEQ ID NO 48
 <211> LENGTH: 244
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium botulinum
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(244)
 <223> OTHER INFORMATION: ColG s3a + s3b domains from type B

<400> SEQUENCE: 48

Ile Asn Val Asp Glu Glu Glu Tyr Asn Asp Asp Phe Glu Cys Ala Asn
 1 5 10 15
 Asn Ile Phe Lys Asn Gln Ile Met Ser Gly Asn Leu Asp Ser Ser Asp
 20 25 30
 Lys Cys Asp Thr Phe Ser Phe Asn Ala Leu Ser Ala Gly Thr Ile Asn
 35 40 45
 Val Thr Leu Glu Asn Ser Asn Ser Asp Ser Ser Thr Val Asn Trp Leu
 50 55 60
 Ala Tyr Ser Ser Glu Asp Thr Asp Asn Tyr Ile Gly Tyr Ala Ser Glu
 65 70 75 80
 Asn Asp Gly Asn Lys Phe Ser Gly Lys Phe Lys Val Asn Lys Pro Gly
 85 90 95
 Lys Tyr Tyr Ile Val Ala Tyr Glu Val Asn Gly Ala Asp Ser Lys Tyr
 100 105 110
 Lys Leu Lys Val Asp Gly Asp Ile Glu Asn Thr Ser Glu Ser Lys Pro
 115 120 125
 Glu Asp Lys Glu Glu Ile Lys Glu Glu Ile Asn Asp Asp Ser Phe Asp
 130 135 140
 Ser Ala Thr Lys Ile Lys Ala Asn Ser Thr Ile Thr Asp Thr Leu Asn
 145 150 155 160
 Gly Glu Asp Asn Lys Asp Ile Cys Tyr Phe Asn Val Asn Asn Asn Ser
 165 170 175
 Asp Leu Asn Ile Glu Leu Asn Ser Leu Thr Asn Leu Gly Val Ala Trp
 180 185 190
 Gln Leu Phe Ser Glu Glu Asp Leu Asp Asn Tyr Ile Ala Tyr Gly Ser
 195 200 205
 Gln Ser Gly Asp Ser Ile Val Gly Thr Ala Asn Val Gln Pro Gly Lys
 210 215 220
 Tyr Tyr Leu Leu Ile Tyr Lys Tyr Thr Gln Ala Asp Gly Ser Tyr Thr
 225 230 235 240
 Phe Thr Ile Lys

<210> SEQ ID NO 49
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium tetani
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(239)
 <223> OTHER INFORMATION: ColG s3a + s3b domains from strain E88

<400> SEQUENCE: 49

Asn Val Ile Tyr Glu Lys Glu Asn Asn Asp Ser Phe Asp Lys Ala Asn
 1 5 10 15
 Lys Ile His Lys Asn Gln Ile Val Met Ala Thr Leu Asp Thr Glu Asp

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20					25					30					
Tyr	Arg	Asp	Thr	Phe	Tyr	Phe	Asp	Ala	Leu	Thr	Ser	Gly	Ser	Ile	Asp
	35						40					45			
Ile	Thr	Ile	Glu	Asn	Ile	His	Gly	Asn	Ser	Asp	Ala	Phe	Asn	Trp	Leu
	50					55					60				
Val	Tyr	Asn	Asp	Glu	Asp	Leu	Asn	Asn	Tyr	Ile	Ala	Tyr	Pro	Thr	Lys
	65					70					75				80
Lys	Glu	Asp	Asn	Lys	Leu	Met	Gly	Ser	Phe	Lys	Val	His	Lys	Pro	Gly
			85					90					95		
Arg	Tyr	Tyr	Ile	Leu	Val	Tyr	Lys	Thr	Ser	Leu	Asn	Lys	Val	Asn	Tyr
			100					105					110		
Lys	Leu	Asn	Ile	Ser	Asp	Ala	Thr	Asn	Met	Ala	Pro	Val	Ile	Lys	Lys
		115					120					125			
Ile	His	Glu	Lys	Glu	Asn	Asn	Asp	Ser	Phe	Glu	Thr	Ala	Asn	Lys	Ile
	130					135					140				
Thr	Leu	Asp	Thr	Leu	Val	Leu	Gly	Asn	Leu	Asp	Tyr	Lys	Asp	Val	Ser
	145					150					155				160
Asp	Ile	Tyr	Ser	Phe	Asp	Ile	Glu	Asn	Thr	Lys	Asp	Leu	Asn	Ile	Lys
			165					170					175		
Leu	Thr	Asn	Leu	Asn	Asn	Leu	Gly	Ile	Ala	Trp	Asn	Leu	Tyr	Lys	Glu
		180					185					190			
Ser	Asp	Leu	Asn	Asn	Tyr	Ile	Ala	Tyr	Gly	Ala	Lys	Ser	Asp	Asn	Ala
		195					200					205			
Ile	Val	Gly	Lys	Cys	Asn	Leu	Ser	Pro	Gly	Lys	Tyr	Tyr	Leu	Tyr	Val
	210					215					220				
Tyr	Lys	Tyr	Ser	Gly	Asp	Lys	Gly	Asn	Tyr	Ser	Val	Ile	Ile	Asn	
	225					230					235				

<210> SEQ ID NO 50
 <211> LENGTH: 243
 <212> TYPE: PRT
 <213> ORGANISM: Clostridium perfringens
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(243)
 <223> OTHER INFORMATION: ColG s3a + s3b domains

<400> SEQUENCE: 50

Glu	Val	Ile	Asn	Glu	Ser	Glu	Pro	Asn	Asn	Asp	Phe	Glu	Lys	Ala	Asn
1			5					10					15		
Gln	Ile	Ala	Lys	Ser	Asn	Met	Leu	Val	Lys	Gly	Thr	Leu	Ser	Glu	Glu
		20					25					30			
Asp	Tyr	Ser	Asp	Lys	Tyr	Tyr	Phe	Asp	Val	Ala	Lys	Lys	Gly	Asn	Val
	35						40				45				
Lys	Ile	Thr	Leu	Asn	Asn	Leu	Asn	Ser	Val	Gly	Ile	Thr	Trp	Thr	Leu
	50				55					60					
Tyr	Lys	Glu	Gly	Asp	Leu	Asn	Asn	Tyr	Val	Leu	Tyr	Ala	Thr	Gly	Asn
	65				70				75					80	
Asp	Gly	Thr	Val	Leu	Lys	Gly	Glu	Lys	Thr	Leu	Glu	Pro	Gly	Arg	Tyr
		85					90						95		
Tyr	Leu	Ser	Val	Tyr	Thr	Tyr	Asp	Asn	Gln	Ser	Gly	Thr	Tyr	Thr	Val
		100					105					110			
Asn	Val	Lys	Gly	Asn	Leu	Lys	Asn	Glu	Val	Lys	Glu	Thr	Ala	Lys	Asp
		115					120				125				
Ala	Ile	Lys	Glu	Val	Glu	Asn	Asn	Asn	Asp	Phe	Asp	Lys	Ala	Met	Lys
	130					135					140				

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Val Asp Ser Asn Ser Lys Ile Val Gly Thr Leu Ser Asn Asp Asp Leu
 145 150 155 160

Lys Asp Ile Tyr Ser Ile Asp Ile Gln Asn Pro Ser Asp Leu Asn Ile
 165 170 175

Val Val Glu Asn Leu Asp Asn Ile Lys Met Asn Trp Leu Leu Tyr Ser
 180 185 190

Ala Asp Asp Leu Ser Asn Tyr Val Asp Tyr Ala Asn Ala Asp Gly Asn
 195 200 205

Lys Leu Ser Asn Thr Cys Lys Leu Asn Pro Gly Lys Tyr Tyr Leu Cys
 210 215 220

Val Tyr Gln Phe Glu Asn Ser Gly Thr Gly Asn Tyr Ile Val Asn Leu
 225 230 235 240

Gln Asn Lys

<210> SEQ ID NO 51
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Brevibacillus brevis
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(239)
 <223> OTHER INFORMATION: ColG s3a + s3b domains

<400> SEQUENCE: 51

Asp Gly Ser Glu Thr Glu Gln Pro Asn Pro Asn Pro Glu Glu Ser Ser
 1 5 10 15

Leu Ser Leu Gly Lys Pro Ile Thr Gly Ile Ile His Pro Gln Lys Pro
 20 25 30

Ser Gln Glu Phe Arg Leu Asp Val Lys Ser Ala Gln Gln Leu Gln Val
 35 40 45

Glu Met Glu Thr Lys Gln Gly Asp Gly Val Ala Trp Leu Val Phe His
 50 55 60

Glu Ala Asp Arg Glu Asn Tyr Ile Ser Tyr Pro Thr Lys Arg Glu Gly
 65 70 75 80

Asn Lys Leu Ile Gly Ser Phe Asp Ala Lys Pro Gly Thr Tyr Tyr Val
 85 90 95

Thr Ala Tyr Thr Tyr Arg Thr Glu Gln Glu Asp Gln Pro Phe Arg Leu
 100 105 110

Leu Val Thr Gly Glu Asp Arg Pro Gln Glu Gln Leu Tyr Gln Glu Asn
 115 120 125

Glu Ser Asn Asp Ser Thr Glu Gln Ala Asn Gly Pro Leu Lys Ile Gly
 130 135 140

Thr Thr Val Ser Gly Asp Met Lys Gly Asn Asp Trp Gln Asp Ile Phe
 145 150 155 160

Ala Phe Gln Val Asp Lys Pro Glu Glu Ile Arg Ile Ser Leu Asn Pro
 165 170 175

Gln Glu Gly Gln Gly Val Thr Trp Met Leu Phe His Glu Gly Asn Leu
 180 185 190

Asp Gln Pro Val Thr Tyr Pro Gln Glu Arg Glu Gly Asn Leu Gln Ser
 195 200 205

Ala His Tyr Gln Val Lys Pro Gly Arg Tyr Phe Leu Tyr Val Tyr Lys
 210 215 220

Tyr Gln Asn Glu Asp Ile Val Tyr Thr Val Glu Thr Lys Gln Arg
 225 230 235

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<210> SEQ ID NO 52
<211> LENGTH: 242
<212> TYPE: PRT
<213> ORGANISM: Paenibacillus dendritiformis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(242)
<223> OTHER INFORMATION: ColG s3a + s3b domains

<400> SEQUENCE: 52
Pro Asn Ala Asp His Glu Pro Asn Asp Ser Trp Glu Gln Ala Val Pro
1          5          10          15
Leu Asp Gly Thr Gly Val Pro Val Ser Gly Lys Leu Ser Asp Thr Asp
20        25        30
Arg Val Asp Val Tyr Arg Phe Asp Ala Gly Lys Ala Glu Gln Trp Thr
35        40        45
Ile Glu Leu Glu Thr Glu Gln Ala Gln Ser Val Ala Trp Val Val His
50        55        60
His Glu Ser Asp Leu Asn Asn Tyr Ala Ala Tyr Pro Thr Gln Val Glu
65        70        75        80
Gly Thr Ser Val Ala Gly Ser Val Asp Ala Val Pro Gly Thr Tyr Tyr
85        90        95
Val Tyr Val Tyr Ser Val Gly Asn Gly Glu Gln Ser Tyr Arg Leu Val
100       105       110
Val Gln Pro Gly Thr Thr Gly Gln Glu Gln Glu Pro Glu Leu Pro Pro
115       120       125
Phe Glu Glu Thr Glu Pro Asn Asp Thr Pro Glu Thr Ala Asn Gly Pro
130       135       140
Ile Pro Ala Gly Arg Pro Val Val Gly Thr Leu Asn Gly Ser Asp Lys
145       150       155       160
Gln Asp Val Phe Ile Ile Asp Val Asp Gln Pro Ala Glu Leu Gln Ile
165       170       175
Glu Leu Glu Arg Arg Leu Gly Ser Gly Val Asn Trp Ile Leu Tyr Arg
180       185       190
Glu Gly Asp Thr Asp Arg Pro Leu Leu Tyr Pro Ser Glu Val Glu Gly
195       200       205
Asn Arg Met Ser Gly Gly Phe Ala Ala Glu Ala Gly Arg Tyr His Leu
210       215       220
Tyr Val Tyr Lys Tyr Thr Asp Glu Asp Ile His Tyr Thr Leu Gln Val
225       230       235       240
Gln His

```

```

<210> SEQ ID NO 53
<211> LENGTH: 238
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(238)
<223> OTHER INFORMATION: ColG s3a + s3b domains

<400> SEQUENCE: 53
Pro Lys Gly Ser Gln Glu Val Gly Asn Asp Asp Thr Phe Glu Thr Ala
1          5          10          15
Asn Gly Pro Ile Lys Ile Asn Thr Asn Tyr Ser Gly Asp Leu Ser Asp
20        25        30
Thr Asp Asn Lys Asp Tyr Tyr Tyr Phe Asn Leu Asp Asn Pro Ser Asn
35        40        45

```

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```

Ile Asn Ile Thr Leu Glu Asn Leu Asp Asn Lys Gly Ile Ser Trp Gln
  50          55          60

Leu Phe His Glu Ser Asp Leu Asn Asn Tyr Val Ala Tyr Pro Thr Thr
  65          70          75          80

Ser Gly Ala Ile Leu Asn Gly Asp Tyr Asn Ala Thr Lys Pro Gly Lys
          85          90          95

Tyr Tyr Ile Leu Val Tyr Asn His Asp Lys Ser Ile Ala Asn Tyr Asn
      100          105          110

Leu Lys Val Asn Phe Gly Asn Asn Asn Asp Asp Gly Val Glu Gln Glu
      115          120          125

Asp Asn Asn Ser Phe Glu Lys Ala Asn Pro Phe Ser Ile Asn Gln Leu
      130          135          140

Val Lys Gly Glu Leu Asp Asn Asn Lys Asp Thr Ser Asp Tyr Phe Lys
      145          150          155          160

Phe Glu Val Lys Glu Asp Ala Gln Leu Asn Ile Ser Leu Glu Lys Thr
      165          170          175

Glu Gly Asp Gly Val Asn Trp Leu Leu Phe Lys Asp Ser Asp Leu Glu
      180          185          190

Asn Tyr Ile Ala Ser Pro Thr Glu Ser Ile Asp Asn Lys Leu Asn Gly
      195          200          205

Lys Val Asp Leu Lys Val Gly Thr Tyr Tyr Leu Glu Val Tyr Gly Tyr
      210          215          220

Gly Ser Ser Pro Val Lys Tyr Asn Phe Lys Val Thr Pro Asn
      225          230          235

```

```

<210> SEQ ID NO 54
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(6)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 54

```

```

Gly Leu Gly Asn Glu Lys
1          5

```

```

<210> SEQ ID NO 55
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Clostridium sporogenes
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(11)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 55

```

```

Gly Asp Ile Asp Lys Ala Pro Leu Lys Asn Glu
1          5          10

```

```

<210> SEQ ID NO 56
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Clostridium botulinum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(16)
<223> OTHER INFORMATION: ColG domain linker from type B

```

```

<400> SEQUENCE: 56

```

```

Gly Asp Ile Glu Asn Thr Ser Glu Ser Lys Pro Glu Asp Lys Glu Glu

```

-continued

```

1           5           10           15

<210> SEQ ID NO 57
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Clostridium tetani
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(11)
<223> OTHER INFORMATION: ColG domain linker from strain E88

```

```

<400> SEQUENCE: 57

```

```

Asp Ala Thr Asn Met Ala Pro Val Ile Lys Lys
1           5           10

```

```

<210> SEQ ID NO 58
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Clostridium perfringens
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(14)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 58

```

```

Gly Asn Leu Lys Asn Glu Val Lys Glu Thr Ala Lys Asp Ala
1           5           10

```

```

<210> SEQ ID NO 59
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Brevibacillus brevis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(9)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 59

```

```

Gly Glu Asp Arg Pro Gln Glu Gln Leu
1           5

```

```

<210> SEQ ID NO 60
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Paenibacillus dendritiformis
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(14)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 60

```

```

Pro Gly Thr Thr Gly Gln Glu Gln Glu Pro Glu Leu Pro Pro
1           5           10

```

```

<210> SEQ ID NO 61
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Clostridium sordellii
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(7)
<223> OTHER INFORMATION: ColG domain linker

```

```

<400> SEQUENCE: 61

```

```

Phe Gly Asn Asn Asn Asp Asp
1           5

```

```

<210> SEQ ID NO 62

```

-continued

```

<211> LENGTH: 95
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(95)
<223> OTHER INFORMATION: PKD domain (s2b) of ColH

```

```

<400> SEQUENCE: 62

```

```

Pro Glu Ile Lys Asp Leu Ser Glu Asn Lys Leu Pro Val Ile Tyr Met
 1             5             10             15
His Val Pro Lys Ser Gly Ala Leu Asn Gln Lys Val Val Phe Tyr Gly
          20             25             30
Lys Gly Thr Tyr Asp Pro Asp Gly Ser Ile Ala Gly Tyr Gln Trp Asp
          35             40             45
Phe Gly Asp Gly Ser Asp Phe Ser Ser Glu Gln Asn Pro Ser His Val
          50             55             60
Tyr Thr Lys Lys Gly Glu Tyr Thr Val Thr Leu Arg Val Met Asp Ser
65             70             75             80
Ser Gly Gln Met Ser Glu Lys Thr Met Lys Ile Lys Ile Thr Asp
          85             90             95

```

```

<210> SEQ ID NO 63
<211> LENGTH: 95
<212> TYPE: PRT
<213> ORGANISM: Clostridium histolyticum
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (1)..(95)
<223> OTHER INFORMATION: PKD domain (s2b) of ColH

```

```

<400> SEQUENCE: 63

```

```

Glu Ile Lys Asp Leu Ser Glu Asn Lys Leu Pro Val Ile Tyr Met His
 1             5             10             15
Val Pro Lys Ser Gly Ala Leu Asn Gln Lys Val Val Phe Tyr Gly Lys
          20             25             30
Gly Thr Tyr Asp Pro Asp Gly Ser Ile Ala Gly Tyr Gln Trp Asp Phe
          35             40             45
Gly Asp Gly Ser Asp Phe Ser Ser Glu Gln Asn Pro Ser His Val Tyr
          50             55             60
Thr Lys Lys Gly Glu Tyr Thr Val Thr Leu Arg Val Met Asp Ser Ser
65             70             75             80
Gly Gln Met Ser Glu Lys Thr Met Lys Ile Lys Ile Thr Asp Pro
          85             90             95

```

```

<210> SEQ ID NO 64
<211> LENGTH: 215
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - PKD-CBD fusion protein

```

```

<400> SEQUENCE: 64

```

```

Glu Ile Lys Asp Leu Ser Glu Asn Lys Leu Pro Val Ile Tyr Met His
 1             5             10             15
Val Pro Lys Ser Gly Ala Leu Asn Gln Lys Val Val Phe Tyr Gly Lys
          20             25             30
Gly Thr Tyr Asp Pro Asp Gly Ser Ile Ala Gly Tyr Gln Trp Asp Phe
          35             40             45
Gly Asp Gly Ser Asp Phe Ser Ser Glu Gln Asn Pro Ser His Val Tyr
          50             55             60

```

-continued

Thr Lys Lys Gly Glu Tyr Thr Val Thr Leu Arg Val Met Asp Ser Ser
 65 70 75 80
 Gly Gln Met Ser Glu Lys Thr Met Lys Ile Lys Ile Thr Asp Pro Val
 85 90 95
 Tyr Pro Ile Gly Thr Glu Lys Glu Pro Asn Asn Ser Lys Glu Thr Ala
 100 105 110
 Ser Gly Pro Ile Val Pro Gly Ile Pro Val Ser Gly Thr Ile Glu Asn
 115 120 125
 Thr Ser Asp Gln Asp Tyr Phe Tyr Phe Asp Val Ile Thr Pro Gly Glu
 130 135 140
 Val Lys Ile Asp Ile Asn Lys Leu Gly Tyr Gly Gly Ala Thr Trp Val
 145 150 155 160
 Val Tyr Asp Glu Asn Asn Asn Ala Val Ser Tyr Ala Thr Asp Asp Gly
 165 170 175
 Gln Asn Leu Ser Gly Lys Phe Lys Ala Asp Lys Pro Gly Arg Tyr Tyr
 180 185 190
 Ile His Leu Tyr Met Phe Asn Gly Ser Tyr Met Pro Tyr Arg Ile Asn
 195 200 205
 Ile Glu Gly Ser Val Gly Arg
 210 215

<210> SEQ ID NO 65
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 65

Gly Pro Leu Gly Ile Ala Gly Pro
 1 5

<210> SEQ ID NO 66
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 66

Gly Pro Gln Gly Ile Ala
 1 5

<210> SEQ ID NO 67
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 67

Gly Pro Gln Gly Ile Leu
 1 5

<210> SEQ ID NO 68
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 68

-continued

Gly Pro Gln Gly Leu Ala
1 5

<210> SEQ ID NO 69
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 69

Gly Pro Gln Gly Leu Leu
1 5

<210> SEQ ID NO 70
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 70

Gly Pro Leu Gly Ile Ala
1 5

<210> SEQ ID NO 71
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 71

Gly Pro Leu Gly Ile Leu
1 5

<210> SEQ ID NO 72
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 72

Gly Pro Leu Gly Leu Ala
1 5

<210> SEQ ID NO 73
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 73

Gly Pro Leu Gly Leu Leu
1 5

<210> SEQ ID NO 74
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic - MMP cleavage sequence

<400> SEQUENCE: 74

-continued

Gly Pro Arg Gly Leu Gln
1 5

<210> SEQ ID NO 75
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - MMP cleavage sequence
 <400> SEQUENCE: 75

Gly Pro Thr Gly Leu Ala
1 5

<210> SEQ ID NO 76
 <211> LENGTH: 58
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - single-stranded oligonucleotide
 encoding collagenase cleavage site
 <400> SEQUENCE: 76

aattagcggc ggaggttcag gtcctctggg aatcgcaggt ccgtccggcg gaggtagc 58

<210> SEQ ID NO 77
 <211> LENGTH: 54
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic - single-stranded oligonucleotide
 encoding collagenase cleavage site
 <400> SEQUENCE: 77

gtacacctcg ccggacggac ctgcgattcc cagaggacct gaacctccgc cget 54

What is claimed:

1. A collagen-binding agent comprising SEQ ID NO:1.
2. A pharmaceutical composition comprising the collagen-binding agent of claim 1 and a pharmaceutically acceptable carrier.
3. A method of treating a condition, the method comprising:
 administering the collagen-binding agent of claim 1 to a subject in an amount effective to treat the condition, wherein the condition is a wound.

4. The method of claim 3, wherein the subject is a mammal.

40 5. A biomedical device comprising a coating that comprises the collagen-binding agent of claim 1, wherein the biomedical device is a stent and wherein the coating further comprises a biodegradable polymer or a collagen.

45 6. A method of using the biomedical device of claim 5 to treat a condition, the method comprising: administering the device to a subject having the condition, wherein the condition is a wound.

* * * * *